Computers and Geotechnics 195 (2026) 108049

Contents lists available at ScienceDirect

Computers and Geotechnics

journal homepage: www.elsevier.com/locate/compgeo

Research Paper

Inter-particle friction effects on submarine landslide dynamics: LBM-DEM simulations of low-aspect-ratio submerged granular collapse

Shu Zhou [a,b,1], Yandong Bi [d,e,1], Xiaolin Tan [b], Zhen Guo [b], Chongqiang Zhu [b], Jin Sun [b], Yu Huang [b,c,*]

[a] Key Laboratory of Mountain Hazards and Engineering Resilience, Institute of Mountain Hazards and Environment, Chinese Academy of Sciences, Chengdu 610299, China
[b] Department of Geological and Hydraulic Engineering, College of Civil Engineering, Tongji University, Shanghai 200092, China
[c] State Key Laboratory of Disaster Reduction in Civil Engineering, Tongji University, Shanghai 200092, China
[d] Shandong Provincial Key Laboratory of Marine Environment and Geological Engineering, Ocean University of China, Qingdao 266100, China
[e] College of Environmental Science and Engineering, Ocean University of China, Qingdao 266100, China

A R T I C L E I N F O

Keywords:
Submarine landslides
Granular column collapses
LBM-DEM
Inter-particle friction

A B S T R A C T

To better understand the influence of particle-scale characteristics on the dynamic processes of submarine landslides, we use a submerged granular column collapse model with a low initial aspect ratio ($a \leq 2$) to conduct a series of coupled lattice Boltzmann method–discrete element method simulations, considering inter-particle friction and the initial aspect ratio. The dynamic characteristics of the particle flow are investigated via an analysis of the runout distance, kinetic energy evolution, and particle contact features. Both the inter-particle friction and aspect ratio are found to significantly influence the dynamic processes of submarine landslides. The runout distance of particles decreases exponentially with increasing inter-particle friction. Compared with a dry system, the runout distance is higher when submerged particles are dominated by fluid viscous resistance; however, the runout distances under dry and submerged conditions are similar when the particles are dominated by frictional stress. Inter-particle friction affects the submerged particle flow via a mechanism in which, as the inter-particle friction increases, the directions of the strong force chains in the system tend to concentrate in the vertical direction, leading to a reduction in the number of strong force chains driving the particle flow. In addition, the densimetric Froude number exponentially increases as the normalized runout distance increases, which could provide a reasonable explanation for the long runout distance of submarine landslides. In general, our findings may have important implications and guiding significance for submarine landslide hazard mitigation, as well as providing an understanding of the mechanisms of complex dynamic processes of two-phase granular particle flow.

1. Introduction

Submarine landslides are common geological hazards that develop on the margins of continental slopes (Hampton et al. 1996; Masson et al. 2006; Tappin et al. 2014; Yincan et al 2017). They can be triggered by earthquakes, waves, and/or currents and can then move down along the slope (Locat and Lee 2002; Talling et al. 2007; Mosher et al. 2010; Vanneste et al. 2013). Because of their extremely large volumes and high flow velocities (Schnyder et al. 2016; Karstens et al. 2023), submarine landslides often have significant disaster impacts, posing a threat to coastal communities and offshore infrastructure, such as communication cables, oil pipelines, and offshore drilling platforms (Hsu et al. 2008; Guo et al. 2019). Consequently, research concerning submarine landslides is very active in the fields of marine geology and coastal engineering.

Different from subaerial landslides, a unique characteristic of submarine landslides is their extremely high mobility, even on gentle slopes of 1–2°. For example, the Storegga submarine landslide, which is the largest in the world, had a runout distance of more than 400 km (Elverhoi et al. 2010; Urlaub et al. 2015; Karstens et al. 2023). Studies of

* Corresponding author at: Department of Geotechnical Engineering, College of Civil Engineering, Tongji University, Shanghai 200092, China.
E-mail address: yhuang@tongji.edu.cn (Y. Huang).
[1] These authors contributed equally to this work.

https://doi.org/10.1016/j.compgeo.2026.108049
Received 4 September 2025; Received in revised form 30 January 2026; Accepted 4 March 2026
Available online 14 March 2026

the Baiyun slide in the South China Sea indicate that it had a runout distance of more than 250 km on an average slope of 0.65° (Ren et al. 2019; Li et al. 2019). Other studies have observed similar extremely high mobility in submarine landslides (Ilstad et al. 2004a; Stevenson et al. 2018; Deng et al. 2018; Schulten et al. 2019). The mechanism of this unique extremely high mobility in submarine landslides has always attracted scientific attention. Recent studies on submarine landslides via large flume tests have shown that the extremely high mobility of submarine landslides is related to hydroplaning effects (Mohrig et al. 1998, 1999; Ilstad et al. 2004a; Sawyer et al. 2012; Du et al. 2022). During the dynamic processes of submarine landslides, the dynamic pressure generated by the sliding causes the flow head to uplift, forming a water film underneath the flow head (Mohrig and Marr 2003; De Blasio et al. 2004; Ilstad et al. 2004b). This process lubricates the base of the flow head and reduces the shear stress on the bed, ultimately leading to an increase in the mobility of submarine landslides (Mohrig et al. 1999). The dynamic pressure generated by submarine landslides and the ability of the flow head to maintain its shape are identified as key factors controlling hydroplaning (Marr et al. 2001; Mohrig and Marr 2003; Ilstad et al. 2004b).

The results from large flume tests have greatly enriched our understanding of the high mobility of submarine landslides. However, these studies are primarily macroscopic investigations. Particle-scale research could obtain more detailed and quantitative descriptions of submarine landslides and therefore contribute to our understanding of the mechanism of the high mobility of submarine landslides (Jing et al. 2018; Sun et al. 2021). Accordingly, some studies have simplified the movement of submarine landslides into immersed granular column collapse, providing an explanation for the extremely mobility of submarine landslides at the particle scale (Rondon et al. 2011; Bougouin and Lacaze 2018; Yang et al. 2020; Lacaze et al. 2021). Rondon et al. (2011) revealed the effect of initial packing on submarine landslides by measuring the excess pore pressure. They found that loose packing results in a positive pore pressure inside the column, leading to a thin and long deposit pattern. Meanwhile, in dense-packing cases, a negative pore pressure and short runout distance are observed. This result indicates that the dilatant and contractive behaviors strongly affects the dynamic processes of submarine landslides. Yang et al. (2020) supplemented the experiments of Rondon et al. (2011) by using a lattice Boltzmann method (LBM) and discrete element method (DEM) coupled simulation. They further found that a strong contact force network in a dense system results in creeping failure behavior. Meanwhile, rapid and catastrophic failure is attributed to liquefaction induced by the positive excess pore pressure of loose packing. On the basis of computational fluid dynamics and DEM coupled simulations of different initial height and column width ratios, Jing et al., (2018) found that eddies in water could carry considerable fluid inertia, significantly modifying the deposit morphology. Collapses under different Stokes numbers were conducted to investigate the influence of ambient water on submarine landslide processes. Studies found that the particle movement gradually changes from being dominated by sliding to being dominated by suspension as the Stokes number decreases (Jing et al., 2019; Lacaze et al., 2021; Sun et al., 2020, 2021). Other studies have investigated the effects of the slope (Kumar et al. 2017), the non-Newtonian flow behavior of pore water (Bougouin et al. 2017), sediment concentration and bedforms (Lu et al. 2024), and the cohesion of particles (Zhu et al. 2022) on the mobility of submarine landslides.

Although the above studies have enhanced our understanding of the dynamic processes of submarine landslides at the particle scale, field investigations have shown that sediments consist of particles that have different sizes, shapes, and mineral compositions, resulting in complex particle characteristics (Blott and Pye 2008; Li et al. 2016; Seiphoori et al. 2021). These complex characteristics contribute to the roughness of the particles in submarine landslides, which is primarily characterized by variations in inter-particle friction (Vo and Nguyen-Thoi 2020). Despite the influence of inter-particle friction on the rheology of dry granular flows and suspensions having been studied (Lootens et al. 2005; Moon et al. 2015; Sivadasan et al. 2019), the influence of inter-particle friction on the dynamics of submarine landslides has not yet been thoroughly investigated.

To fill this scientific gap, this paper uses the immersed granular column collapse model to study the influence of inter-particle friction (The following text is referred to as “particle friction”) on the dynamic processes of submarine landslides at the particle scale. The rest of this paper is organized as follows. The basic theories of the coupled LBM-DEM method, the corresponding validation, and model configuration of the study are described in Section 2. The simulation results of the runout, energy evolution, and particle contact characteristics are analyzed in Section 3. Section 4 analyzed the flow region and hydroplaning using dimensionless numbers. Based on the findings of this study, several key points are discussed in Section 5. Some of the critical findings are then summarized in Section 6.

2. Numerical methodology

The fluid dynamics for the simulation of submerged particle collapse is resolved using LBM and DEM is used to describe the particle motion. The coupling of LBM-DEM is implemented via additional collision terms in LBM and additional force and torque terms in DEM.

2.1. Lattice Boltzmann method

Using the Bhatnagar–Gross–Krook (BGK) approximation (Bhatnagar et al. 1954), the governing equation of LBM can be written as (Succi 2001)

$$f_i(\mathbf{x}+\mathbf{c}_i\Delta t, t+\Delta t) = f_i(\mathbf{x}, t) + \Omega_i^{BGK} \tag{1}$$

where Δt is the time step and $f_i(\mathbf{x}, t)$ is a discrete particle distribution function at time t positioned at $\mathbf{x}$ pointing in the i-th direction at each lattice node. In this study, a D3Q27 lattice structure is used for the three-dimensional LBM simulations to achieve good accuracy and efficiency (Guo and Shu, 2013; Krüger et al., 2017). $\mathbf{c}_i$ is a pre-defined lattice velocity, and Ω_i^{BGK} is the collision term, which can be expressed as

$$\Omega_i^{BGK} = \frac{\Delta t}{\tau}\left(f_i^{eq}(\mathbf{x}, \mathbf{u}, t) - f_i(\mathbf{x}, t)\right) \tag{2}$$

where $f_i^{eq}(\mathbf{x}, \mathbf{u}, t)$ is the equilibrium distribution function, $\mathbf{u}$ is the macroscopic fluid velocity, and τ is a single relaxation parameter that determines how fast the particle distribution functions recover to the equilibrium state and can be related to the lattice fluid viscosity: $\nu = (\tau - 0.5)/3$.

The macroscopic fluid density ρ and the velocity $\mathbf{u}$ can be obtained by computing the zeroth and first moments of the particle distribution function.

$$\begin{aligned} \rho &= \sum_{i=0}^{26} f_i(\mathbf{x}, t) \\ \rho\mathbf{u} &= \sum_{i=0}^{26} f_i(\mathbf{x}, t)\mathbf{c}_i \end{aligned} \tag{3}$$

The macroscopic fluid viscosity ν_f can be obtained via (He and Luo 1997)

$$\nu_f = c_s^2\left(\tau - \frac{1}{2}\right)\frac{\Delta x^2}{\Delta t} \tag{4}$$

where c_s is the speed of sound, equal to $1/\sqrt{3}$ in lattice units; and Δx is the lattice spacing.

2.2. Discrete element method

In DEM, the motion of each particle i can be described by the Newtonian motion equations (Cundall and Strack, 1979; MiDi, 2004):

$$\frac{dv_i}{dt} = g + \frac{1}{m_i}\sum_j F_{ij}^n + F_{ij}^t \tag{5}$$

$$\frac{d\omega_i}{dt} = \frac{1}{I_i}\sum_j x_{ij} \times F_{ij}^t \tag{6}$$

where m_i, I_i, x_i, v_i, and ω_i are the mass, moment of inertia, position, velocity, and rotational velocity of particle i, respectively; g is the gravitational acceleration; and F_{ij}^n and F_{ij}^t are the normal and tangential forces between particles i and j, respectively, and are calculated using the Hertz–Mindlin model in this study (Mindlin 1949; Rojek 2018). Particle friction is considered via the Coulomb yield criterion (Zhu et al. 2020, 2023). The motion equations are explicitly integrated using the second-order velocity-Verlet algorithm (Verlet 1968).

2.3. LBM-DEM coupling

The LBM-DEM coupling is implemented using the partially saturated-cell method (Noble and Torczynski 1998). Considering an additional collision term Ω^s and a weighting function $B(\mathbf{x}, \tau)$ that counts the solid fraction of a lattice node $\mathbf{x}$, the governing equation of LBM (Eq. (2)) can be rewritten as

$$f_i(\mathbf{x} + \mathbf{c}_i\Delta t, t + \Delta t) = f_i(\mathbf{x}, t) + (1 - B(\mathbf{x}, \tau))\Omega_i^{BGK} + B(\mathbf{x}, \tau)\Omega_i^s \tag{7}$$

According to Noble and Torczynski (1998), there are two types of collision terms for the solid phase Ω^s: non-equilibrium bounce back and superposition (SP). However, Najuch and Sun (2019) showed that bounce back fails to correctly capture the stresslet. Meanwhile, SP can accurately calculate both the stresslet and the torque. Therefore, SP is used in this study. For SP, the collision term Ω^s is expressed as

$$\Omega_i^s = f_i^{eq}(\rho, \mathbf{u}_s) - f_i(\mathbf{x}, t) + (1 - \frac{\Delta t}{\tau})\left[f_i(\mathbf{x}, t) - f_i^{eq}(\rho, \mathbf{u})\right] \tag{8}$$

where $\mathbf{u}_s$ is the particle velocity. The weighting function $B(\mathbf{x}, \tau)$ is related to the solid fraction ε_s and the relaxation time τ and can be expressed as

$$B(\mathbf{x}, \tau) = \frac{\varepsilon_s(\tau/\Delta t - 1/2)}{(1 - \varepsilon_s) + (\tau/\Delta t - 1/2)} \tag{9}$$

The solid fraction ε_s can be evaluated using a cell decomposition method (Owen et al. 2011; Najuch and Sun 2019).

The local macroscopic hydrodynamic force $F^H(\mathbf{x})$ can be calculated from the momentum transfer between the fluid and solid domains:

$$F^H(\mathbf{x}) = \frac{\Delta x^d}{\Delta t} B(\mathbf{x}) \sum_{i=0}^{26} \Omega_i^s \mathbf{c}_i \tag{10}$$

where d is the dimension. Hence, the local macroscopic hydrodynamic torque $T^H(\mathbf{x})$ can be calculated via the cross product of the force and the corresponding lever arm:

$$T^H(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_s) \times F^H(\mathbf{x}) \tag{11}$$

where $\mathbf{x}_s$ is the center of the particle. The total hydrodynamic force F^H and torque T^H acting on a particle can be obtained by summing the local values of all of the lattice cells covered by the solid particle. To account for the hydrodynamic influence of the fluid in DEM, the hydrodynamic force and torque are added to the right-hand sides of Eqs. (5) and (6), respectively.

To synchronize the LBM and DEM simulations, the time step between LBM and DEM should satisfy as following:

$$\Delta t_{\mathrm{DEM}} = \frac{\Delta t_{\mathrm{LBM}}}{n} \tag{12}$$

where n is the sub-cycle of the DEM calculation. During the sub-cycling procedure, the values of the hydrodynamic force F^H and torque T^H acting on a particle are assumed to remain constant. Note that the LBM time step is determined by Eq. (4), using the predefined fluid viscosity, the relaxation time, and the spatial resolution. Meanwhile, the DEM time step should satisfy the critical time step, which can be determined by the time for a Rayleigh wave to propagate along the average particle radius (Li et al. 2005; Shen et al. 2022).

2.4. Model validation

2.4.1. Single particle sinking in a viscous fluid

A single particle moving in a viscous fluid is a classic example to validate fluid–particle interactions. When a spherical particle with radius r moves in a viscous fluid with constant velocity U, the fluid drag force F_d can theoretically be evaluated as (Zhao et al. 2014)

$$F_d = \frac{1}{2}\pi r^2 \rho_f C_d U^2 \tag{13}$$

where ρ_f is the density of the fluid and C_d is the drag coefficient and depends on the particle Reynolds number, which is related to the particle velocity U, diameter d, and kinematic viscosity of the fluid ν, such that

$$\mathrm{Re} = \frac{|U|d}{\nu} \tag{14}$$

In this study, we used a model consisting of a spherical particle with a diameter of 0.0005 m moving in a simulation domain with a length, width, and height of 0.005 m, 0.005 m, and 0.005 m, respectively (Fig. 1 a). The lattice spacing is 2.5×10^{-5} m. The fluid density is 1000 kg/m^3, and the relaxation time τ is 0.65. Using Eq. (4), we can obtain the kinematic viscosity of the fluid. Therefore, by setting different particle velocities U, we can simulate a particle moving in viscous fluids with different Re values. Equation (13) is used to numerically calculate the drag coefficient and to compare it with experimental results. Fig. 1b shows a comparison of the drag coefficients (C_d) obtained from the LBM-DEM simulation (red squares) and the experimental results of Brown and Lawler (2003) (black dots) under Re values of 0.008, 0.08, 0.8, 8, and 100. Our LBM-DEM results show excellent agreement with the experimental measurements over the Re range from 0.008 to 100. This means that the presented LBM-DEM algorithm has the ability to accurately simulate particles moving in a viscous fluid.

2.4.2. Model verification with the submerged particle column collapse model

To examine the reliability of the LBM-DEM method under dense packing conditions, we validated the algorithm using the immersed granular collapse experiments conducted by Bougouin and Lacaze (2018). In their study, Bougouin and Lacaze (2018) performed a series of immersed granular collapse tests to investigate the flow behavior of particles under free fall, inertial, and viscous regimes. Here, we select the test results from the viscous regime to verify the LBM-DEM algorithm. The dimensions of the simulation domain are 0.3 m × 0.036 m × 0.3 m (length, width, and height, respectively). For the initial particle column geometry, it is set to 0.01 m × 0.036 m × 0.05 m (length, width, and height, respectively), ensuring an aspect ratio of $a = 0.5$. The initial packing fraction is consistent with the experimental value of 0.64. Other simulation model parameters, including fluid and particle properties, match the experimental conditions (Table S1).

The comparison between the LBM-DEM simulation results and the experimental data is presented in Fig. 2. It can be seen that at the three time points of $1/3t_f$, $2/3t_f$, and t_f (where t_f is the time when particles

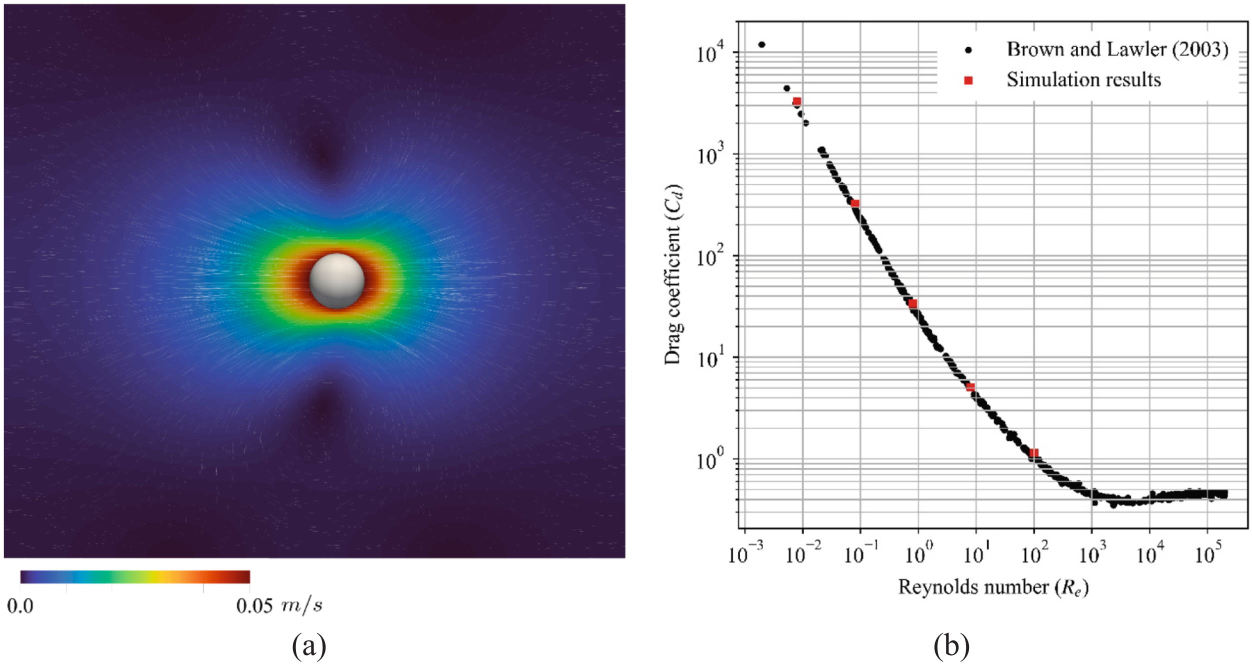


Fig. 1. Validation results. (a) Model setup. (b) Comparison of drag coefficients (C_d) obtained from the lattice Boltzmann method (LBM)-discrete element method (DEM) simulation (red squares) and Brown and Lawler (2003) (black dots) for different Reynolds numbers.

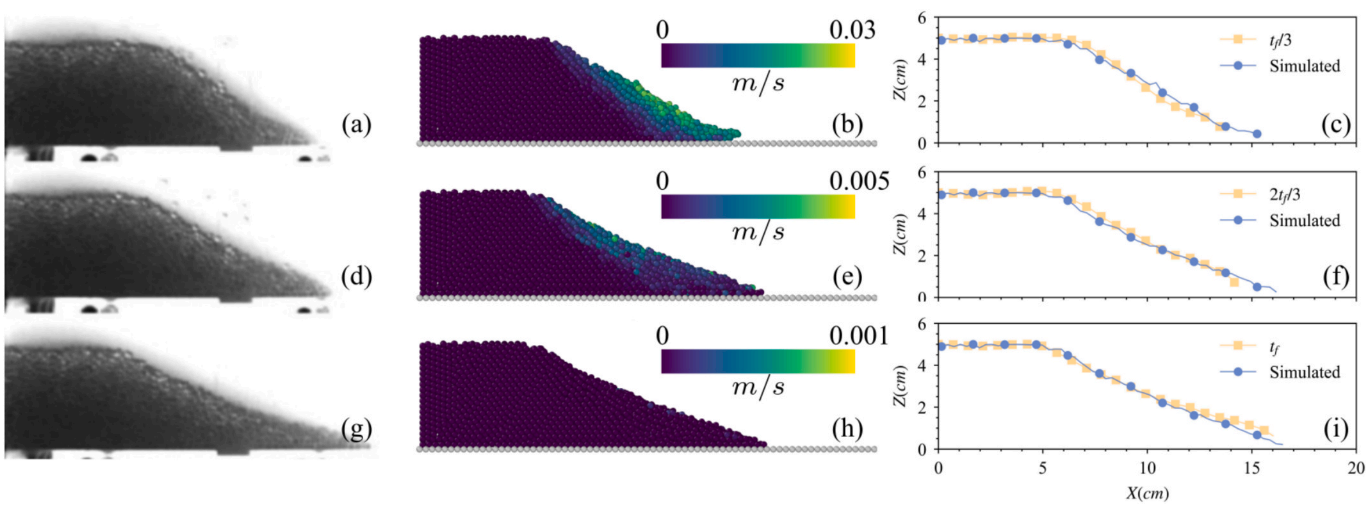


Fig. 2. Comparison of simulation results with the experimental data from Bougouin and Lacaze (2018). The first, second, and third rows show the deposit patterns at 1/3 t_f, 2/3 t_f and t_f, respectively. (t_f represents the time when the particles stop).

stop), the simulation results show good agreement with the experimental data. This demonstrates that the LBM-DEM method used in this study is accurate and reliable for simulating immersed granular collapse.

2.5. Model setup and simulation procedures

The immersed granular column collapse model for the particle-scale dynamic characteristics investigation of submarine landslides is shown in Fig. 3. The granular column has an initial length and width of $L_i = 25$ mm and $l_y = 20$ mm, respectively. Meanwhile, the initial height H_i is determined by the aspect ratio ($a = H_i/L_i$) in the different simulation cases. For our simulation cases with $a = 0.8$, 1.5, and 2, the initial heights are 20 mm, 37.5 mm, and 50 mm, respectively. The spherical particles in the column have a mean diameter of $d_p = 1$ mm with a weak polydispersity of 10% standard deviations following a Gaussian distribution (Yang et al. 2020). The simulation domain has a length, width, and height of l_x, l_y, and l_z, respectively. In our simulation, l_y is at least 20 times the particle diameter and is maintained at 20 mm. Meanwhile, to reduce the simulation consumption while meeting the need for a sufficient space for particle motion, l_x and l_z vary with the aspect ratio a. l_x is set to 160 mm, 200 mm, and 240 mm for $a = 0.8$, 1.5, and 2, respectively, and l_z is set to 30 mm, 60 mm, and 75 mm for $a = 0.8$, 1.5, and 2, respectively. At the bottom of the simulation domain, a rough bed that consists of particles with a uniform size of $d_p = 1$ mm is generated to maintain a certain basal roughness.

Periodic boundary conditions are used in the y-direction for both the fluid and particles to eliminate solid boundary effects. For the fluid simulation, the top surface is set to free-slip boundary conditions, while no-slip boundary conditions are defined for the rest of the walls. For the particle simulation, except for the walls in the y-direction, the walls are set to free-slip boundary conditions. The simulation parameters for both the fluid and particles are summarized in Table 1. Except for the

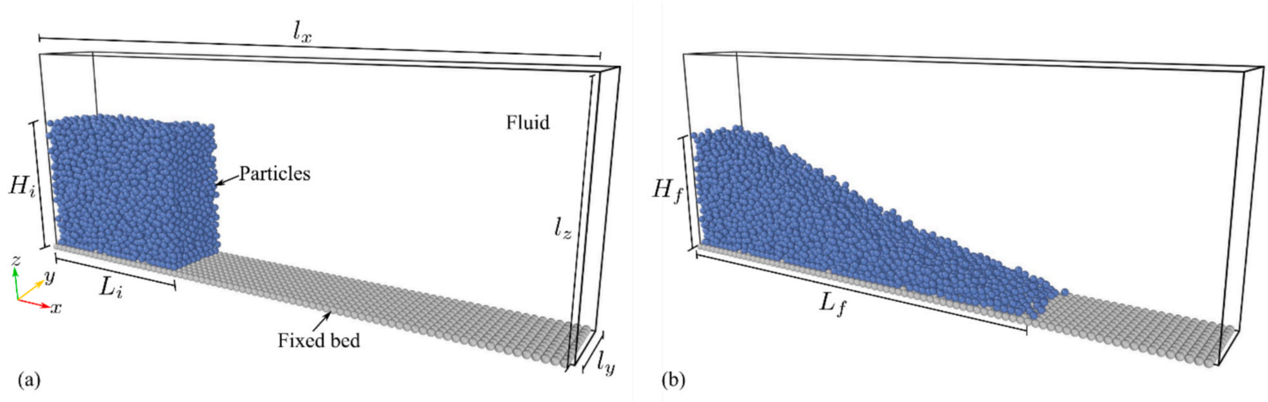


Fig. 3. Sketches showing the configuration of granular column collapse in a viscous fluid. (**a**) Initial state. (**b**) Final state. H_i and L_i are the initial height and length of the granular column, respectively; l_x, l_y, and l_z represent the length, width, and height of the simulation domain, respectively; and L_f and H_f are the final runout distance and deposit height, respectively.

Table 1
Input parameters for the simulation cases. DEM indicates the discrete element method, and LBM indicates the lattice Boltzmann method.

DEM parameter	Value	LBM parameter	Value
Mean Particle diameter d_p (mm)	1	Fluid density ρ_f (kg/m^3)	1000
Particle density ρ_p (kg/m^3)	2500	Lattice spacing Δx (mm)	0.1
Young's modulus E (Pa)	10^9	Relaxation time τ	0.65
Poisson's ratio ν	0.24	Time step Δt (s)	5×10^{-5}
coefficient of restitution e	0.65	Dynamic viscosity μ_f (Pa•s)	0.01
coefficient of friction μ_p	0.1 ~ 1	DEM-LBM coupling sub-cycles n	100
Time step dt (s)	5×10^{-7}	Lattice cells per particle diameter m	10

coefficient of friction of the particle being set to 0.1–1 with an interval of 0.1 in the different simulation cases, the DEM simulation parameters use the recommended values of Yang et al. (2020) for LBM-DEM simulations. The dynamic viscosity of the fluid is 0.01 Pa•s, which was determined using Eq. (4) with a relaxation time of $\tau = 0.65$, lattice spacing of $\Delta x = 0.1$ mm, and time step of $\Delta t = 5 \times 10^{-5}$. The number of lattice cells per particle diameter m is 10, which is sufficient to satisfy the simulation accuracy when using SP with a relaxation time of $\tau = 0.65$ according to Najuch and Sun (2019). The Stokes number St is 0.753, which can be evaluated such that (Bougouin and Lacaze 2018)

$$St = \frac{1}{18\mu_f}\sqrt{\frac{\rho_p(\rho_p - \rho_f)gd_p^3}{2}} \tag{15}$$

According to the phase diagram, which consists of St and the density ratio $r = \sqrt{\rho_p/\rho_f} = 1.58$ (Jing et al. 2019; Sun et al. 2021), the granular flow regime of this study is located in the weak viscous regime.

The simulation procedures are conducted via the following steps. (1) Particles with a reduced mean size equal to 0.1 d_p are generated (randomly distributed in space but not overlapping with each other) in a box with dimensions of L_i, l_y, and 1.2H_i, with the location of the box overlapping the initial area of the collapsed column. All particles are then gradually enlarged to the targeted d_p. (2) The particles are subjected to an effective gravity of $g' = g(\rho_p - \rho_f)/\rho_p$ to make the particles settle (g is the gravitational acceleration and its value is 9.81 m/s^2). During this process, we used trial-and-error to modify the friction coefficient μ_p and the rolling friction coefficient μ_r to obtain the target initial volume fraction ϕ_0 of the granular column ($\phi_0 = 0.583 \pm 0.003$). When the kinetic energy of the system is less than 2×10^{-8} J, the initial collapsed columns are considered to be stationary and the settling is terminated. The particles above the target initial height are removed, and additional steps are run to eliminate the influence of the removed particles. (3) The LBM-DEM coupling procedures are implemented, and the granular column is then released by removing the release gate. During this process, the effective gravity is used to consider the buoyancy of the fluid. After the release, the granular column collapses and spreads horizontally. The simulation time for the collapse is 2 s, which is sufficiently long for the particles to nearly stop. Note that we checked that the kinetic energy of the system was less than 2×10^{-8} J for all simulation cases.

2.6. *Post-processing*

The dynamic processes of solid–fluid mixtures with solid volume fractions ϕ and various grain sizes are dominated by different stresses. For those with high ϕ and large grain sizes, the fluid tends to be dominated by grain collisional and frictional stresses. Conversely, fluid with low ϕ and fine particles tends to be dominated by fluid viscous stresses (Song et al. 2018). Therefore, quantitatively studying the contributions of particle collisional, frictional, and fluid viscous stresses to the macroscopic flow dynamics could help reveal the mechanism of gravity flows (Ng et al. 2013, Ng et al., 2019; Zhou et al. 2019; Song and Choi 2021). The Bagnold number N_B, which is a dimensionless number that describes the ratio of the collisional stresses to the viscous stresses, can be defined as (Bagnold 1954)

$$N_B = \frac{\phi_{ft}\rho_p d_p^2\dot{\gamma}}{(1 - \phi_{ft})\mu_f} \tag{16}$$

where ϕ_{ft} is the average solid volume fraction at the surge front region of a particle and $\dot{\gamma}$ is the shear rate. As described by Yang et al. (2020), the shear rate can be evaluated via the maximum surge front velocity v_{ft} during the entire simulation and the corresponding average depth of the surge front h_{ft}:

$$\dot{\gamma} = \frac{v_{ft}}{h_{ft}} \tag{17}$$

Note that the depth h_{ft} and velocity v_{ft} of the surge front are difficult to determine because the surge front region is unclear and because the moving particle and ambient water do not have a distinct surface. Here, we refer to the method proposed by Yang et al. (2020) for the surge front velocity and height calculation. The specific process can be found in the supplementary materials.

The Savage number N_S describes the relative importance of the grain collisional to frictional stresses without considering the fluid contribution (Savage 1984), that is,

$$N_S = \frac{\rho_p d_p^2 \dot{\gamma}^2}{(\rho_p - \rho_f) g h_{ft} \tan\theta} \tag{18}$$

where θ is the internal friction angle. In this study, the value of $\tan\theta$ is equal to the coefficient of friction μ_p (Yang et al. 2021a).

To characterize the ratio of the frictional to fluid viscous stresses, the Friction number N_F is defined as the ratio of N_B to N_S (Iverson 1997):

$$N_F = \frac{\phi_{ft}(\rho_p - \rho_f) g h_{ft} \tan\theta}{(1 - \phi_{ft}) \mu_f \dot{\gamma}} \tag{19}$$

Another important dimensionless number that characterizes the dynamic processes of submarine landslides is the densimetric Froude number Fr_d. This number describes the ratio of the hydrodynamic pressure to the total vertical load on the flow head and can be defined as (Mohrig et al. 1998)

$$Fr_d = \frac{v_{ft}}{\sqrt{(\rho_p/\rho_f - 1) g h_{ft} \cos\beta}} \tag{20}$$

where β is the inclination angle. Because this study uses a configuration with a 0° bed inclination, the value of $\cos\beta$ is 1.

Previous studies have identified critical values for different flow regimes for the above-mentioned dimensionless numbers. For N_B, Bagnold (1954) reported that collisional stresses start to dominate over fluid viscous stresses when $N_B > 200$. $N_S > 0.1$ indicates that collisional stresses dominate over frictional stresses (Savage 1984; Iverson 1997). The critical value of N_F is still controversial. For subaerial debris flows, Iverson (1997) reported that frictional stresses dominate over fluid viscous stresses for $N_F > 2000$. Meanwhile, Parsons et al. (2001) reported that frictional stresses dominate when $N_F > 100$. The numerical result of Yang et al. (2020) indicates that a critical value of $N_F = 200$ could describe the immersed particle collapse model well. Our model configuration is similar to that of Yang et al. (2020); therefore, the critical value of N_F was set to 200 in this study. Previous studies have shown that the onset of hydroplaning has a critical Fr_d value between 0.3 and 0.4 (Mohrig et al. 1998; Marr et al. 2001; Ilstad et al. 2004b; Sun et al. 2020). Hydroplaning could be experienced by flows with Fr_d values greater than the critical value.

3. Numerical Results

The purpose of the following analysis is to show the influence of the aspect ratio and the particle friction coefficient on the dynamic characteristics of granular collapse and to further illustrate insights into submarine landslides. Several critical parameters obtained from this study are summarized in Table 2.

3.1. Dynamic collapse processes

Snapshots of the flow for different aspect ratios at 0.05, 0.25, 0.5, and 2 s are shown in Fig. 4. The overall dynamic characteristics are similar for the columns with different aspect ratios. After the gate is removed, the bottom particles start to move, leading to the collapse of particles at the top-right corner, which is a sign of movement. Meanwhile, particles in the lower corner remain nearly static (dead zone), a shear plane is formed, and the entire granular column retains a nearly rectangular shape (first column in Fig. 4). As the collapse process develops, the particles come into contact with the bed and begin to spread horizontally in the x-direction (second column in Fig. 4). As the energy of the particles dissipates in the spreading processes, moving particles

Table 2
Summary of the kinematic measurements and dimensionless values for the LBM-DEM simulations.

a	ϕ_O	μ_p	L_f*	H_f*	v_{ft} (mm/s)	h_{ft} (mm)	ϕ_{ft}	N_B	N_S	N_F	Fr_d
$a = 0.8$	0.581	0.1	1.582	0.747	92.79	4.90	0.523	5.231	0.131	42.716	0.347
		0.2	1.398	0.860	77.32	5.10	0.518	4.082	0.040	109.829	0.283
		0.3	1.280	0.901	70.50	5.32	0.516	3.540	0.020	194.826	0.253
		0.4	1.243	0.924	65.76	4.91	0.506	3.439	0.016	225.823	0.245
		0.5	1.205	0.936	62.90	5.26	0.509	3.113	0.010	348.191	0.227
		0.6	1.163	0.940	57.61	4.62	0.497	3.103	0.011	342.668	0.223
		0.7	1.145	0.943	57.97	5.09	0.503	2.901	0.007	478.938	0.213
		0.8	1.197	0.945	58.36	5.40	0.505	2.766	0.005	620.960	0.208
		0.9	1.158	0.943	57.86	5.46	0.506	2.713	0.004	720.107	0.205
		1.0	1.191	0.945	59.18	4.87	0.496	2.996	0.005	607.283	0.222
$a = 1.5$	0.584	0.1	4.058	0.499	196.63	6.51	0.525	8.407	0.247	35.388	0.637
		0.2	3.540	0.573	179.57	6.32	0.516	7.646	0.114	70.716	0.591
		0.3	3.134	0.606	165.28	6.30	0.505	6.823	0.068	110.505	0.546
		0.4	2.952	0.634	151.80	6.28	0.504	6.194	0.042	159.194	0.501
		0.5	2.856	0.638	148.43	6.42	0.501	5.878	0.030	210.245	0.485
		0.6	2.690	0.648	141.20	6.65	0.494	5.237	0.021	279.317	0.454
		0.7	2.716	0.658	140.00	6.59	0.497	5.287	0.018	321.816	0.451
		0.8	2.616	0.658	135.60	6.75	0.499	5.044	0.013	403.372	0.432
		0.9	2.595	0.668	131.25	6.83	0.505	4.928	0.011	490.889	0.416
		1.0	2.650	0.657	137.42	6.50	0.495	5.239	0.013	456.913	0.447
$a = 2.0$	0.586	0.1	5.926	0.411	244.15	6.54	0.515	10.070	0.382	27.726	0.790
		0.2	5.114	0.472	226.00	6.59	0.510	9.064	0.160	59.651	0.728
		0.3	4.578	0.510	209.21	6.39	0.508	8.592	0.101	89.574	0.685
		0.4	4.202	0.524	204.57	6.10	0.500	8.473	0.082	108.603	0.685
		0.5	4.012	0.541	189.97	6.45	0.500	7.466	0.049	163.616	0.619
		0.6	3.874	0.548	180.79	6.39	0.497	7.090	0.038	199.991	0.592
		0.7	3.664	0.551	178.28	6.75	0.496	6.572	0.027	263.769	0.569
		0.8	3.644	0.556	176.55	6.65	0.496	6.642	0.025	297.477	0.568
		0.9	3.644	0.555	177.47	6.42	0.498	6.935	0.024	310.612	0.580
		1.0	3.643	0.559	171.38	6.46	0.490	6.475	0.020	353.031	0.559

Note that only the averaged front properties of the flow and the averaged dimensionless values are listed. L_f* and H_f * are the normalized final runout distance and top height, respectively.

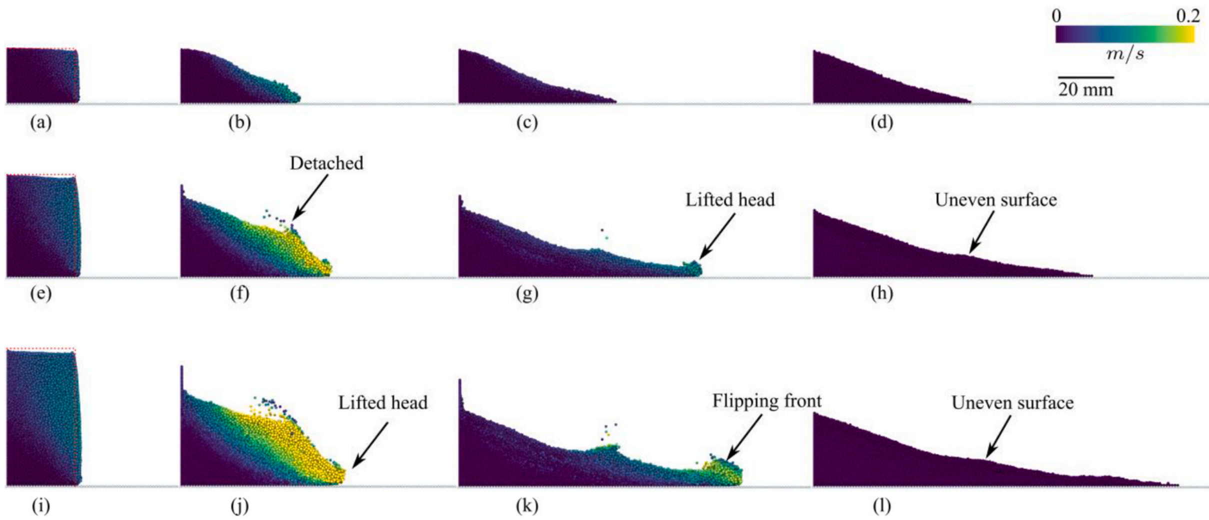


Fig. 4. Collapse sequences of granular columns obtained with $\mu_p = 0.4$. The rows present cases with different aspect ratios: $a = 0.8$ (top row); $a = 1.5$ (middle row); and $a = 2.0$ (bottom row). The columns present snapshots at different times: $t = 0.05$ s (first column); $t = 0.25$ s (second column); $t = 0.5$ s (third column); and $t = 2$ s (fourth column). Particle colors indicate the velocity magnitude. The red dotted lines indicate the initial profiles of the granular columns.

slow and are deposited (third and fourth columns in Fig. 4). The above-mentioned processes agree with experimental observations (Rondon et al. 2011; Bougouin and Lacaze 2018; Sun et al. 2021).

Differences in the particle motion are observed for columns with different aspect ratios. For $a = 0.8$, the collapse time is relatively short and there is no obvious particle detachment near the water–particle interface (WPI). With increasing a, we clearly observe particle detachment near the WPI (Fig. 4f); however, there is also a noticeable lifting of the frontal head of the moving particles (Fig. 4g and 4j). These two phenomena are related to the velocity increase of the moving particles. As the particle velocity increases, the particle Re also increases, which results in an increase in the drag force acting on the particles. Meanwhile, a collapse with a higher a generates a larger eddy at the WPI (Yang et al. 2020), which could also increase the drag force. Given these two contributions, particles at the WPI will detach from the main body. This detachment has also been observed in flume tests of submarine landslides (Ilstad et al. 2004b; Du et al. 2023). The uplifted frontal head indicates the occurrence of hydroplaning (Mohrig et al. 1998). When the moving particles have high velocities, the dynamic pressure underneath the frontal head is also high (Liu et al. 2020). The pressure difference between the bottom and top of the particle layer causes the flow head to lift, inducing hydroplaning (Marr et al. 2001; Mohrig and Marr 2003).

The deposit pattern in cases with $a = 0.8$ has a triangular shape with a flattened surface. This deposit form is similar to the result of Rondon et al. (2011) obtained from a case with $a = 0.5$. With increasing a, although the deposit maintains a nearly triangular shape, the surface is uneven (Fig. 4h and 4l). The generation of an uneven deposit for cases with high a could be attributable to (1) the influence of eddies and/or (2) the flipping front induced by hydroplaning. For the former, as previously mentioned, high a cases generate large eddies with higher energy that drag the moving particles and therefore reshape the deposit form (Jing et al. 2018; Yang et al. 2021b). When submarine landslides experience hydroplaning, the frontal head will uplift and flip (Fig. 4k). The flipped flow head will increase the flow depth behind the frontal head, resulting in an uneven deposit form (the frontal head has a higher depth, as shown in Fig. 4k). Experimental results confirm that the flipping front not only influences the dynamic processes of submarine landslides but also the deposit form (Ilstad et al. 2004a).

The initial collapse is a key stage in understanding the dynamic characteristics of the overall particle collapse. To investigate the influence of μ_p on the initial collapse, we used a normalized velocity v^* to describe the movement of the particles at the initial stage. The normalized velocity v^* is defined as the ratio of the particle velocity to the averaged velocity $\bar{v}$ in cases with $\mu_p = 0.1$ at a counterpart aspect ratio a. Spatial distributions of the normalized velocity v^* of the particles at $t = 0.05$ s are shown in Fig. 5. The lower-left corners for all cases have a small v^* (deep blue area), indicating that the particles remain nearly static; these areas can be identified as a dead zone. The spatial distribution of the dead zone indicates that the failure is initiated with the sliding of particles at the top-right corner. Because the boundary of the dead zone is more obvious and continuous than that of other zones, the boundary line of the dead zone can be regarded as the shear plane. For all cases, the shear planes extend through the toe of the column. This phenomenon has been observed in previous studies (Jing et al. 2018, 2019; Sun et al. 2021), and Yang et al. (2020) found that it was related to the initial volume fraction of the particle column. A comparison of the results obtained from different μ_p indicates that the area of the dead zone increases and v^* decreases as μ_p increases for all a, such that the number of moving particles and the intensity of the collapse decrease. This trend is similar to that for dry granular flow (Lai et al. 2023); however, the collapse is still sliding-dominated at high μ_p ($\mu_p = 0.9$) and not toppling-dominated, as is found in dry systems. This result may be related to the fact that the lubrication and buoyancy of water reduce the friction between particles (Ness and Sun 2015).

3.2. Runout characteristics

The entire propagation of the granular column collapse can be divided into stages of (I) acceleration, (II) constant velocity, and (III) deceleration. To analyze these stages, the normalized frontal head position L^* and the normalized top height H^*, which are common choices in the literature (Lajeunesse et al. 2005) are extracted such that

$$L^* = (L - L_i)/L_i \tag{21}$$

$$H^* = H/H_i \tag{22}$$

where L and H are the frontal flow position and the top height of the collapsed column, respectively. The time is scaled by $\sqrt{h_i/g}$, a choice that has been verified in previous studies (Meruane et al. 2010), where h_i is the corresponding initial height of the column. Another important parameter to describe runout processes is the propagation time, which refers to the time taken by stage II. Here, we followed Yang et al., (2021b) to defined the propagation time t_{run} as the difference between the deposit time t_{95} (the time that 95% of the final runout distance is

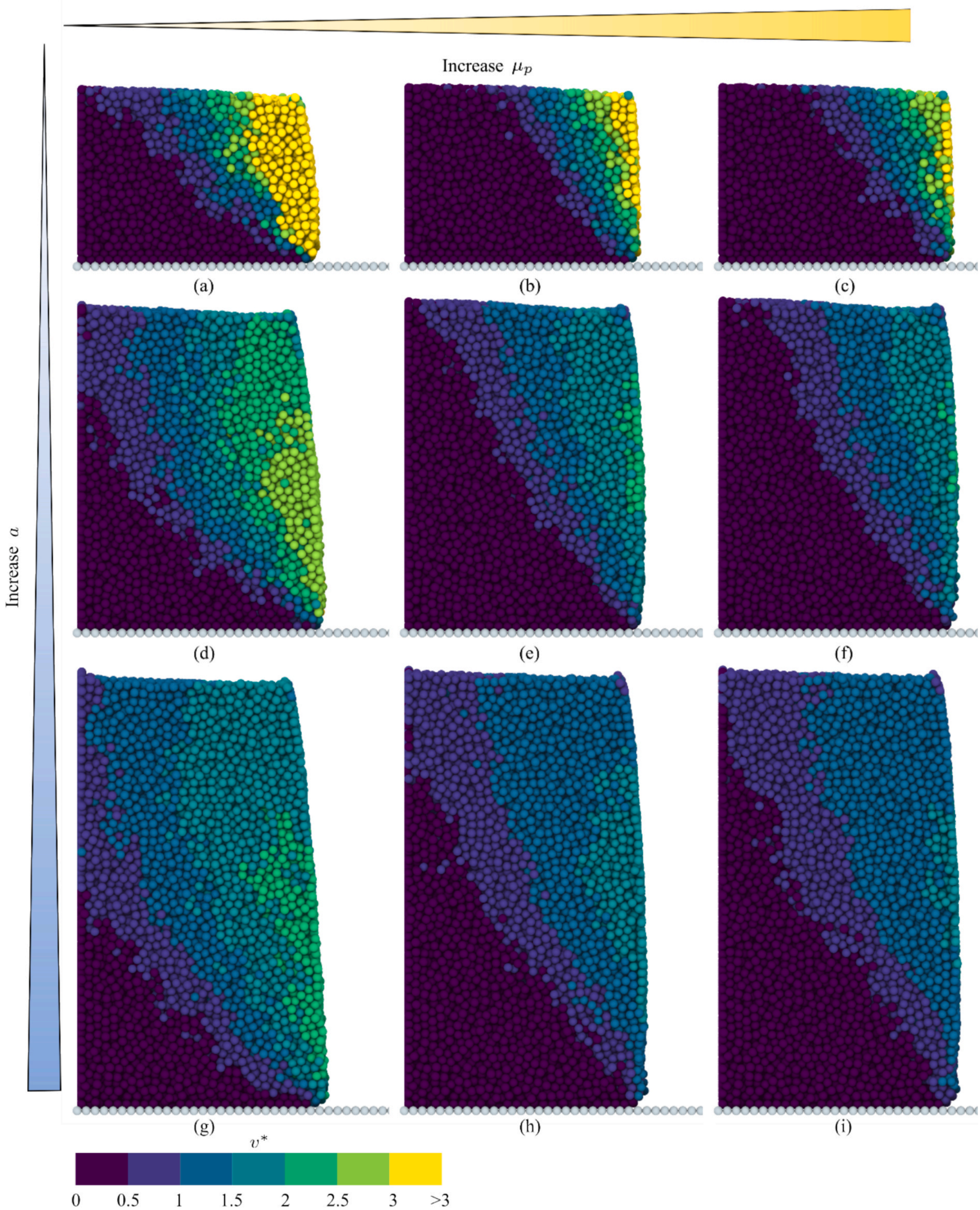


Fig. 5. Spatial distributions of the normalized velocity v^* of particles for collapse at different a and μ_p at $t = 0.05$ s. The rows contain cases with different aspect ratios: $a = 0.8$ (top row); $a = 1.5$ (middle row); and $a = 2.0$ (bottom row). The columns indicate $\mu_p = 0.1$ (first column), $\mu_p = 0.5$ (second column), and $\mu_p = 0.9$ (third column). The color indicates the magnitude of v^*.

reached) and the triggering time t_{tr} (the interception of the fitted straight line of the front position in stage II with the time axis).

The temporal evolution of L^* obtained from cases with different a and μ_p is shown in Fig. 6a. The propagating velocity (the slope of the curves) at stage II increases as a increases; however, it tends to decrease as μ_p increases. Meanwhile, consistent with previous studies (Jing et al. 2018; Sun et al. 2021; Yang et al. 2021b), the duration of collapse for each a becomes similar after the distance and time are normalized, having a value of approximately 8–9. However, variations in μ_p for the same a also yield discernible differences, corresponding to a decrease in the duration of collapse as μ_p increases. To quantify this difference, the physical t_{run} and t_{tr} are plotted in Fig. 6b. With increasing μ_p, t_{tr} gradually increases, indicating that the particle friction coefficient is a controlling factor for the initial collapse. Note that previous studies have reported

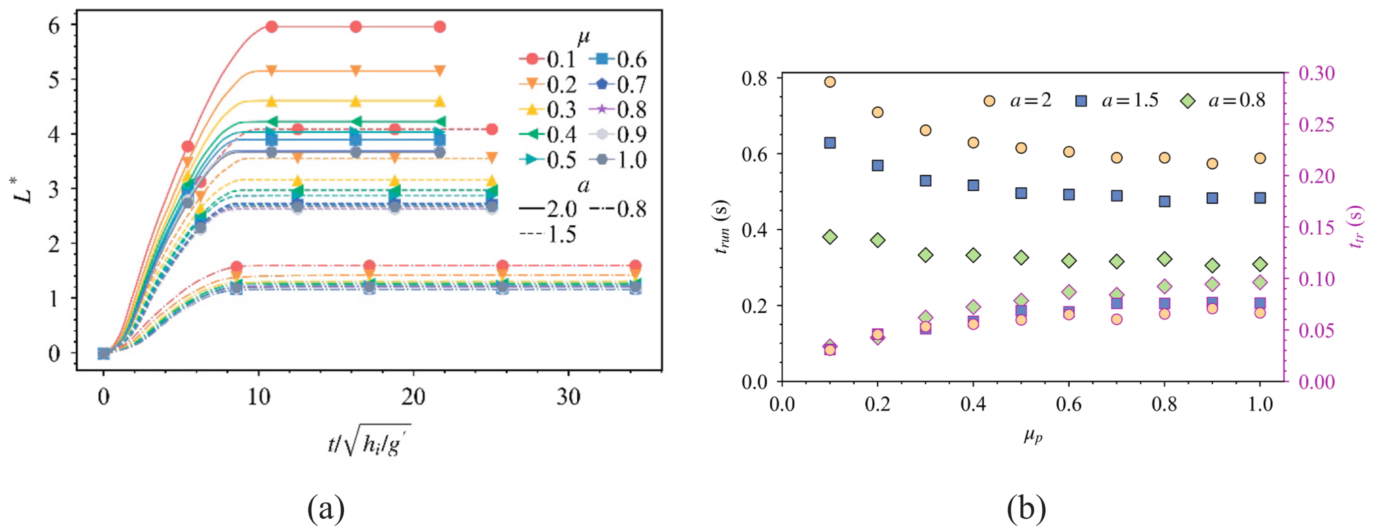


Fig. 6. Temporal evolution of the normalized frontal head position. **(a)** Normalized frontal head position versus normalized time. The dash-dotted lines represent $a = 0.8$, the dotted lines represent $a = 1.5$, and the solid lines represent $a = 2$. The colors distinguish cases with different μ_p (see legend). The time is normalized by $\sqrt{h_i/g'}$, where h_i is the corresponding initial height of the column. **(b)** Propagation time t_{run} and triggering time t_{tr} versus μ_p. The time is in physical units. Note: The dots with black edges represent t_{run}, corresponding to the y-axis on the left; The dots with pink edges represent t_{tr}, corresponding to the y-axis on the right.

that a lower particle friction coefficient can generate a denser column that has a slower collapse process (Rondon et al. 2011; Izard et al. 2018; Yang et al. 2020). Therefore, our result only indicates that the reduction of μ_p at $\phi_0 \approx 0.58$ reduces t_{tr}. For submarine landslides composed of low-friction particles but with high ϕ_0, the contributions of ϕ_0 and μ_p to the initial collapse require further investigation. Another interesting finding is that, for cases with $\mu_p = 0.1$ and 0.2, t_{tr} is nearly the same for all a; meanwhile, for other cases, t_{tr} decreases as a increases. This result depends on whether the particles are dominated by frictional viscous or fluid viscous stresses (see Section 4). An analysis of t_{run} shows that t_{run} decreases as μ_p increases and increases as a increases. The relationship between t_{run} and a agrees well with previous results (Staron and Hinch 2005; Jing et al. 2018), and the values of t_{run} are close to the experiment results of Yang et al., (2021b) under the corresponding a condition. Note that the trend of t_{run} decreasing as μ_p increases becomes less pronounced, which implies that, when μ_p exceeds a critical value, the contribution of μ_p to the particle movement becomes less significant.

The normalized final runout distance L_f* and the top height H_f * are shown in Fig. 7a and 7b, respectively. L_f* decreases as a power law form of μ_p for all of the a values that we studied. Meanwhile, different decreasing trends of L_f* are observed under different a values, i.e., the decrease trend for $a = 0.8$ is less pronounced than those for $a = 1.5$ and 2. The above results may be related to the critical a_c, which has been found in previous studies for dry and submerged particle collapse (Bougouin and Lacaze 2018; Jing et al. 2018; Yang et al. 2021b; Lai et al. 2023). When $a < a_c$, the relationship between L_f* and a is linear. When $a > a_c$, they have a power-function relationship. The influence of the critical a_c was also found for different μ_p in a previous study of dry particle collapse, i.e., $L_f^* = 1.204(\mu_p)^{-0.291}a$ when $a < a_c = 1$ and $L_f^* = 1.349(\mu_p)^{-0.245}a^{0.68}$ when $a > a_c = 1$ (Lai et al. 2023). Our observations of the relationship between μ_p and L_f* under different a conditions are similar to the findings of the above-mentioned study; however, the quantitative relationship between L_f*, μ_p, and a for the immersed particle collapse requires further research. In addition, in Fig. 7a, we also provide the flow distances of the dry particles corresponding to μ_p and a. It can be seen that when the particles have a small value of μ_p, the L_f* value of the submerged particles is higher than that of the dry particles. However, when μ_p is relatively higher, the runout distances are quite similar for submerged and dry cases. The explanation for this phenomenon will be discussed in Section 4.2 in conjunction with the flow regimes of the particles. The relationship between H_f *, μ_p, and a shows that H_f * decreases as a increases but increases with μ_p (Fig. 7b). The

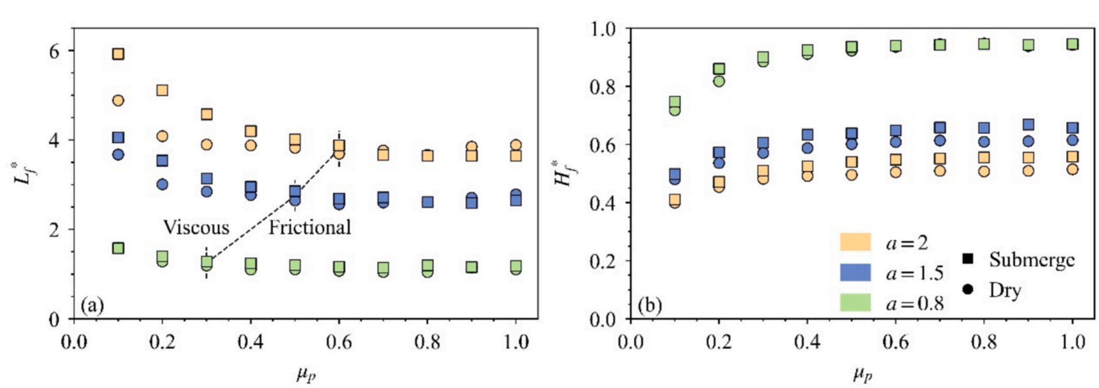


Fig. 7. Plots of the particle friction coefficient μ_p versus **(a)** the normalized final runout distance L_f* and **(b)** the normalized top height H_f *. Note: Regarding the classification of viscous and frictional flow regimes in Fig. 7 (a), please refer to Section 4.2. The meaning of the word 'viscous' is that the flow is dominated by fluid viscous stress, and the meaning of the word 'frictional' is that the flow is dominated by particle frictional stresses. Meanwhile, the explanation for the similarity in the runout distance of submerged and dry particles under high friction conditions can be found in Section 4.2.

effect of μ_p on H_f * with columns with different a differs. For columns with $a = 0.8$, as μ_p increases, H_f * increases to approach 1.0, meaning that some particles remain static during the collapse process. For columns with $a = 1.5$ and 2, H_f * increases with increasing μ_p; however, the value of H_f * is 0.4–0.6, meaning that most of the particles collapse. This result coincides with the result of the initial collapse analysis in Section 3.1.

3.3. Energy evolution

The temporal evolution of the kinetic energy is a critical factor in understanding the dynamic processes of granular collapse (Jing et al. 2018). Here, we used the normalized kinetic energy of the particles $<\overline{E}_p>$ and fluid $<\overline{E}_f>$ to describe the kinetic energy evolution of the system:

$$
\begin{aligned}
<\overline{E}_x> &= \frac{1}{2}\sum_{i=1}^{n_p} m_{p_i}\left|u_{p_i}\right|^2 \Big/ E_0 \\
<\overline{E}_y> &= \frac{1}{2}\sum_{i=1}^{n_p} m_{p_i}\left|v_{p_i}\right|^2 \Big/ E_0 \\
<\overline{E}_z> &= \frac{1}{2}\sum_{i=1}^{n_p} m_{p_i}\left|w_{p_i}\right|^2 \Big/ E_0
\end{aligned}
\tag{23}
$$

$$
<\overline{E}_p> = <\overline{E}_x> + <\overline{E}_y> + <\overline{E}_z> \tag{24}
$$

$$
<\overline{E}_f> = \frac{1}{2}V_{fc}\sum_{i=1}^{n_{fc}} \alpha_{f_i}\rho_{f_i}\left|\mathrm{u}_{f_i}\right|^2 / E_0 \tag{25}
$$

where $<\overline{E}_x>$, $<\overline{E}_y>$, and $<\overline{E}_z>$ are the normalized kinetic energies of the particles in the x-, y-, and z-directions, respectively; u_{p_i}, v_{p_i}, and w_{p_i}

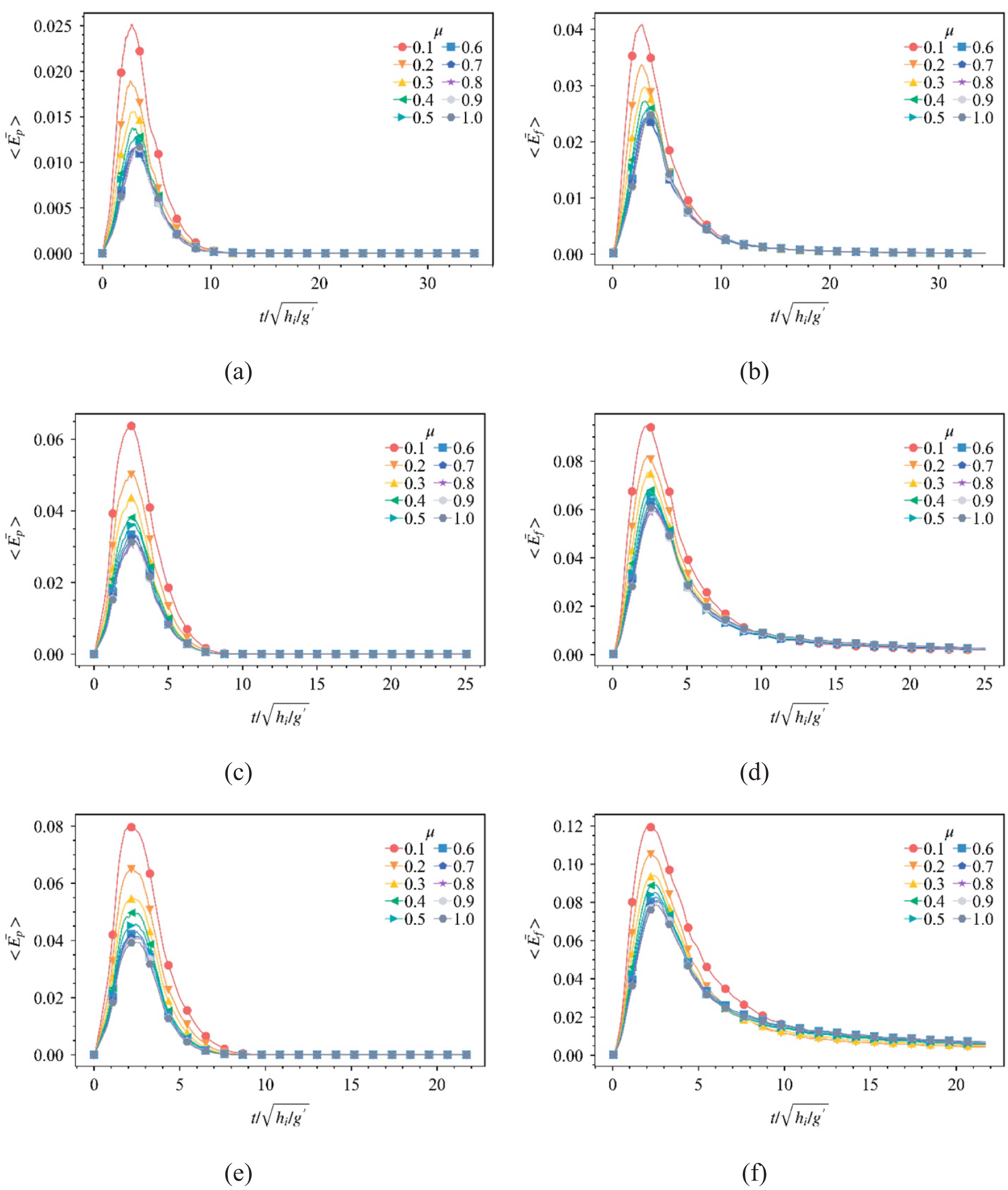


Fig. 8. Temporal evolution of the kinetic energy of the particles (left column) and fluid (right column) for $a = 0.8$ (top row), $a = 1.5$ (middle row), and $a = 2$ (bottom row).

are the velocity components in the *x*-, *y*-, and *z*-directions of the *i*-th particle, respectively; m_{p_i} is the corresponding mass; and n_p is the number of particles. ρ_{f_i}, V_{fc}, α_{f_i}, and u_{f_i} are the fluid density, volume, porosity, and flow velocity of the *i*-th lattice, respectively; n_{fc} is the number of lattices in the simulation domain; and E_0 is the total energy of the system at the initial stage, which can be obtained by calculating the potential and kinetic energies at the initial time $t = 0$ s.

The temporal evolution of the kinetic energy of the particles and fluid extracted from cases with different a and μ_p is shown in Fig. 8. Both $< \overline{E}_p >$ and $< \overline{E}_f >$ increase from zero to a peak value and then decrease over time, which is consistent with the three-stage evolution pattern of granular collapse. The kinetic energy of the fluid is higher than that of the particles for all cases, which may be related to our use of a higher viscosity fluid (the flow regime is located in the weak viscous regime). This is consistent with the study of Jing et al. (2019) in that $< \overline{E}_f >$ in the viscous regime is greater than $< \overline{E}_p >$, while in the inertial regime and the free-fall regime, $< \overline{E}_f >$ is less than $< \overline{E}_p >$. A higher value of a leads to higher kinetic energy for both the fluid and the particles, which is consistent with the earlier-mentioned rapid collapse of the particle column under high a conditions. The rate of dissipation of the fluid kinetic energy is much lower than that of the particle kinetic energy for all cases, and as a increases, the rate of dissipation of the fluid kinetic energy slows. This indicates that the fluid retains significant kinetic energy after the particles stop moving and that the residual kinetic energy of the fluid increases as a increases. The residual fluid kinetic energy is carried primarily by the induced eddies (Yang et al. 2021b). This result confirms our earlier analysis that high a will generate eddies with high energy, leading to uneven deposit surfaces. The influence of μ_p on both $< \overline{E}_p >$ and $< \overline{E}_f >$ shows a similar pattern to that of the influence on the runout distance, that is, the kinetic energy decreases as μ_p increases and the kinetic energy difference between the cases with different μ_p values is smaller for high μ_p.

The normalized peak value of the kinetic energy and the corresponding normalized time (t_{pk}) of both the particles and fluid are plotted in Fig. 9. The values of the normalized peak kinetic energy range from 1% to 12%, and the value of t_{pk} is 2–3.5. Meanwhile, as μ_p increases, the peak kinetic energy generally tends to decrease while t_{pk} increases. This implies that an increase in μ_p may lead to a slower collapse process. Note that the relationship between μ_p and t_{pk} is not monotonic and that the discussion here only concerns the trend. This may be related to the fact that μ_p becomes less significant for the collapse dynamics once it exceeds a critical value. The peak values also indicate that μ_p increases as a increases.

The partial kinetic energies of particles in the *x*- and *z*-directions ($< \overline{E}_x >$ and $< \overline{E}_z >$, respectively) are shown in Fig. 10. According to the energy evolution characteristics in the vertical and horizontal directions, the granular column collapse can be divided into collapse, heap, and spread phases (Vo and Nguyen 2024). The collapse phase extends from the initial collapse to the peak vertical kinetic energy. The heap phase occurs between the peak vertical kinetic energy and the peak horizontal kinetic energy, demonstrating the transformation of kinetic energy. The spread phase consists of the remaining time after the peak horizontal kinetic energy. From the partial kinetic energy curve, it can be seen that, during the collapse stage, $< \overline{E}_z >$ rapidly increases and reaches its peak under the action of gravity while $< \overline{E}_x >$ remains small, indicating that the particle movement is primarily vertical. The influence of μ_p on $< \overline{E}_z >$ mainly manifests in two aspects: the peak value and the time at which the peak occurs. The peak value of $< \overline{E}_z >$ tends to decrease as μ_p increases, and the time that the peak $< \overline{E}_z >$ occurs tends to be delayed. Meanwhile, $< \overline{E}_z >$ increases as a increases. Note that, as a increases, $< \overline{E}_z >$ disappears earlier, which suggests that the horizontal motion dominates the system in the spreading phase under high a conditions. After the collapse phase, distinguished by the peak $< \overline{E}_z >$, the horizontal kinetic energy $< \overline{E}_x >$ dramatically increases and reaches its peak value. The peak value of $< \overline{E}_x >$ is heavily dependent on the ability of the particle collapse to transfer kinetic energy from the vertical direction to the horizontal direction and is thought to be an indicator of the overall mobility of the immersed granular collapse (Topin et al. 2012). The peak value of $< \overline{E}_x >$ decreases as μ_p increases and increases as a increases, indicating a decrease in the mobility of the column. This relationship agrees well with those between $L_f{}^*$, μ_p, and a. To quantify the efficiency of the energy conversion from the vertical to horizontal directions, we calculated the ratio of the peak $< \overline{E}_x >$ to the peak $< \overline{E}_z >$ (E_x^{max}/E_z^{max}), as plotted in Fig. 10d. E_x^{max}/E_z^{max} has a maximum value at $a = 1.5$, indicating a non-linear relationship with a. This result agrees with previous studies and can be explained by the cushioning effect of the fluid, the contact reducing effect of interstitial fluids, and the drag effect of the eddies (Jing et al. 2018; Yang et al. 2021b). The influence of μ_p on E_x^{max}/E_z^{max} is nonlinear, meaning that, as μ_p increases, E_x^{max}/E_z^{max} nonlinearly decreases. This may be because, at high μ_p, the friction between particles results in a significant loss of kinetic energy and therefore leads a low E_x^{max}/E_z^{max}. Note that the lower friction coefficient of the particles can more effectively convert the particle kinetic energy from the vertical direction to the horizontal direction, explaining the increased mobility observed in our simulations with low μ_p.

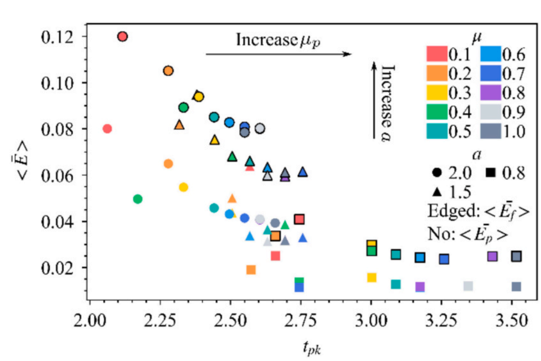


Fig. 9. Relationship between the normalized peak values of both the particle and fluid kinetic energies and the corresponding normalized time. The colors indicate cases with different μ_p. The shape of the symbol indicates cases with different a. The kinetic energy of the particles is outlined in black, while the kinetic energy of the fluid is not.

3.4. Particle contact characteristics

The particle–particle contacts determine the column collapse characteristics and help explain the microscopic features of the collapse. The contact force networks in the selected cases that have $\mu_p = 0.1$, 0.5, and 0.9 at $t = 0.05$ s are shown in Fig. 11. The spatial distribution of the force chains indicates that lower positions have stronger force chains. It is these strong force chains that support the entire granular column and prevent the particles from moving. At $a = 0.8$, the strong force chains tend to be concentrated in the lower right corner, while at $a = 1.5$ and 2, they are primarily concentrated in the lower left corner. The orientation of the force chain changes as a increases. The differences in the spatial distribution of the force chain for different a values may be related to the different initial failure modes, as mentioned earlier. With increasing μ_p, we observe two phenomena: the strength of the force chain increases and the orientation of the contact forces appears to shift. To quantitatively describe the spatial distribution variation of the force chain network with μ_p and a, we extracted the strong contact force chains, those that primarily carry the load and can be defined as $F_c > F_{cc}$ (where F_{cc} is the critical contact force distinguishing strong and weak force chains). Here, F_{cc} is defined as three times the average contact force, as suggested by Yang et al. (2020), to better quantify strong force chains in immersed granular collapse. The average orientations of the strong contact force chains projected into the *xy*-plane are shown in Fig. 12.

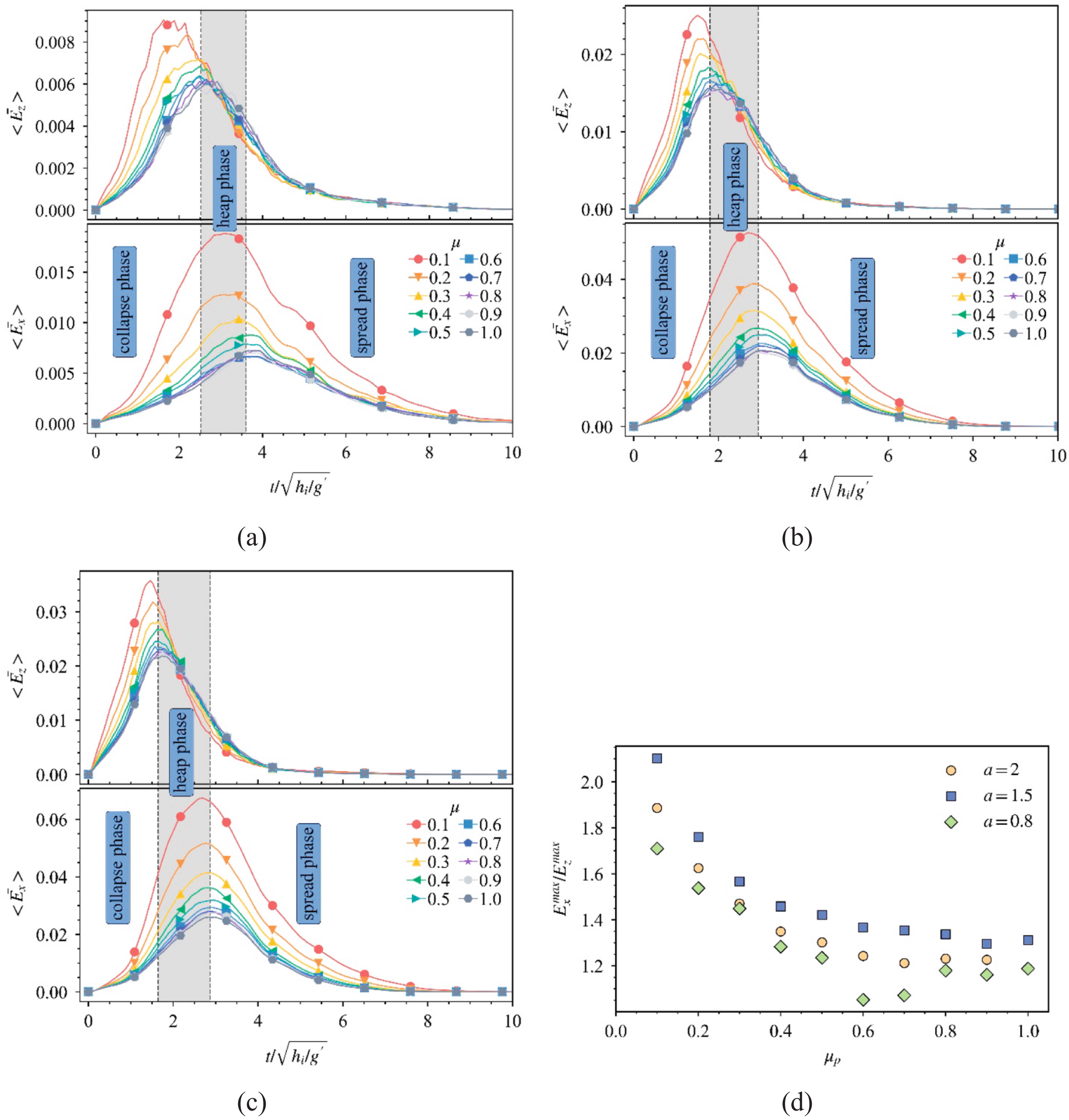


Fig. 10. Evolutions of the normalized kinetic energy of the particles in the vertical and horizontal directions: **(a)** $a = 0.8$; **(b)** $a = 1.5$; and **(c)** $a = 2$. **(d)** Ratio of the peak vertical to horizontal kinetic energy.

The average orientation of the strong contact force chain in the upper right direction (0–90°) tends to increase as μ_p increases and that in the upper left direction (90–180°) tends to decrease as μ_p increases. This indicates that the strong force chains, which mainly carry the load, orientate close to vertical as μ_p increases. Therefore, a force chain network that tends toward the vertical direction leads to a smaller driven force as μ_p increases, causing the initial collapse process to be relatively slow at higher μ_p. The orientations of the strong force chain are nearly the same at $a = 1.5$ and 2, indicating that these two cases have similar collapse processes; however, the magnitude of the strong contact force at $a = 2$ is higher than that at $a = 1.5$, resulting in a high driven force at $a = 2$ and leading to a faster collapse during the initial stage. In addition, the trigger time is shorter at $a = 2$ than at $a = 1.5$ (Fig. 6b).

To quantify the anisotropy of the granular contacts during the collapse, the fabric tensor of the contact is defined as

$$C_{ab} = \frac{15}{2}\left(\phi_{ab} - \frac{1}{3}Tr(\phi)\delta_{ab}\right) \tag{26}$$

where a and b represent the x-, y-, and z-directions; δ_{ab} is the Kronecker delta function; and the tensor ϕ is defined as

$$\phi_{ab} = \sum_{n=1}^{N_p} \frac{r_a r_b}{r^2} \tag{27}$$

where N_p is the number of pair interactions in the system, r_a is the a-component of the radial vector between two pairwise interacting particles, and r is the magnitude of the radial vector. The magnitude of anisotropy of the granular system can be represented by the second invariant of the tensor C_{ab}:

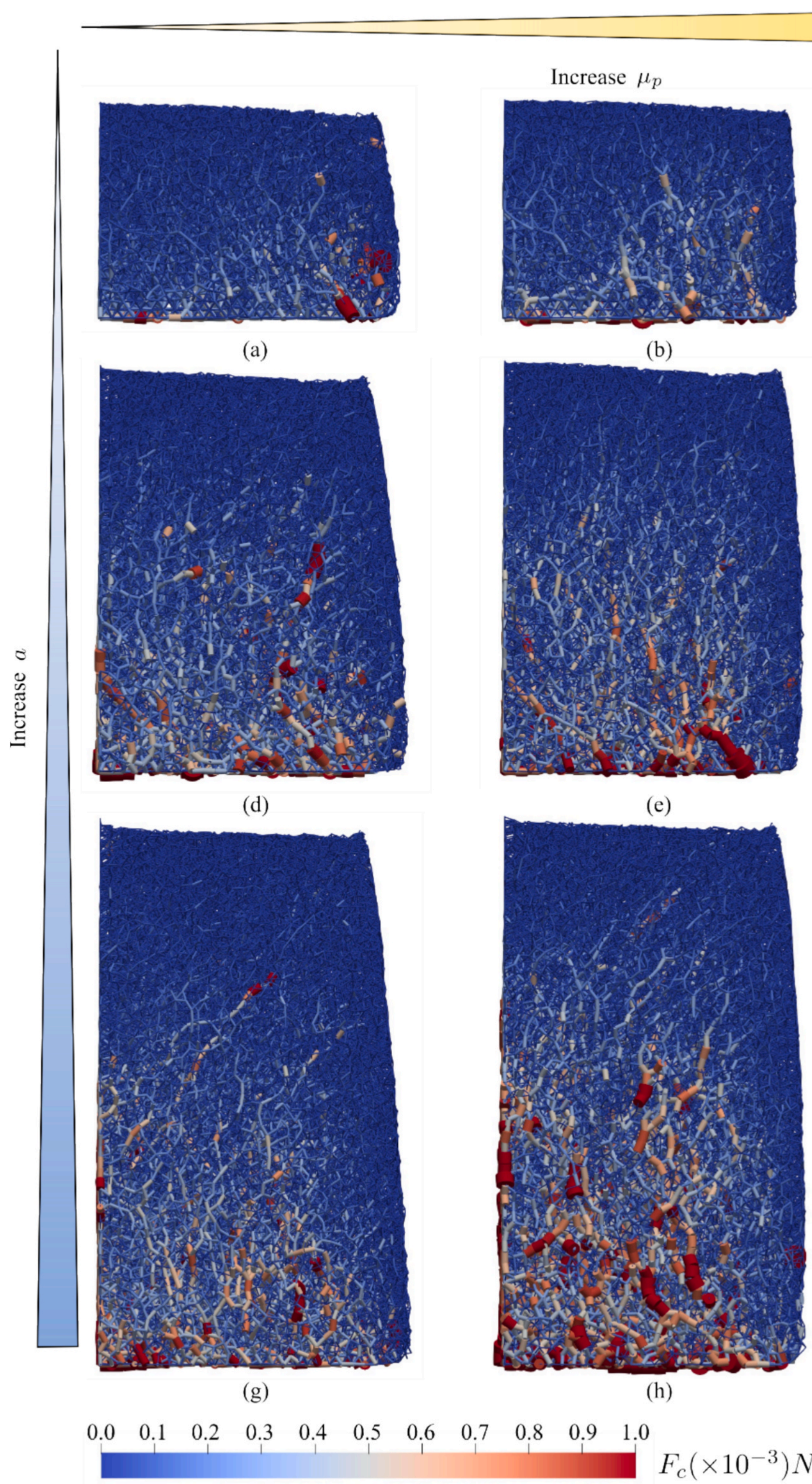


Fig. 11. Comparison of the contact force networks in the immersed granular columns for cases with different a and μ_p at $t = 0.05$ s. The rows present cases with different aspect ratios: $a = 0.8$ (top row); $a = 1.5$ (middle row); and $a = 2.0$ (bottom row). The columns indicate $\mu_p = 0.1$ (first column), $\mu_p = 0.5$ (second column), and $\mu_p = 0.9$ (third column). The color indicates the magnitude of the contact force, and the magnitude of the contact force is proportional to the diameter of the contact segments.

$$a(C_{ab}) = \sqrt{\frac{3}{2}C_{ab}C_{ab}} \tag{28}$$

Fig. 13 shows the temporal evolution of $a(C_{ab})$ during the collapse. When $t = 0.05$ s, the values of $a(C_{ab})$ for $a = 0.8$ with $\mu_p = 0.1$, 0.5, and 0.9 are 0.192, 0.225, and 0.223, respectively, for $a = 1.5$ with $\mu_p = 0.1$, 0.5, and 0.9 are 0.247, 0.308, and 0.308, respectively, and for $a = 2$ with $\mu_p = 0.1$, 0.5, and 0.9 are 0.246, 0.334, and 0.339, respectively. This result generally agrees with the results of the force chain analysis, in that the anisotropy of the system tends to increase as a and μ_p increase. The time series of $a(C_{ab})$ indicates that $a(C_{ab})$ dramatically increases at the beginning of the collapse. This may be due to the rapid adjustment of the force chains within the particles after the gate is removed. When $a(C_{ab})$

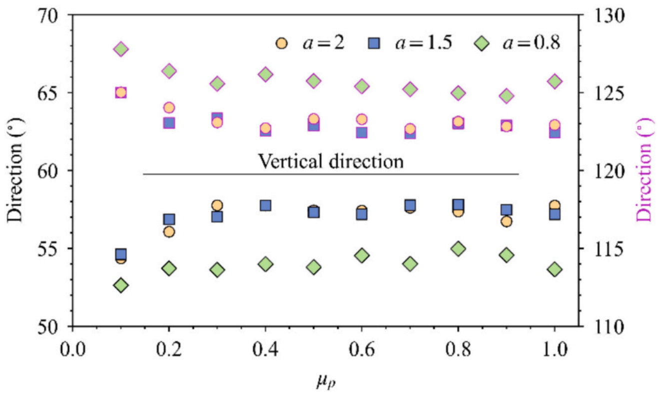


Fig. 12. Averaged orientations of the upper-left and upper-right strong contact force chains.

reaches its peak value, two patterns for different a values exist. For $a = 0.8$, $a(C_{ab})$ gradually decreases with time. Meanwhile, when $a = 1.5$ and 2, $a(C_{ab})$ appears to have a second peak. Comparing the kinetic energy evolution with $a(C_{ab})$, the peak at $a = 0.8$ is seen to occur at approximately the same time as the peak kinetic energy in the x-direction. Meanwhile, the first peak at $a = 0.15$ and 2 corresponds to the linear stage of the kinetic energy increase in the z-direction and the second peak coincides with the peak of the kinetic energy in the x-direction. On the basis of these results, we speculate that the rapid adjustment of the particle contacts after the gate is removed and the resulting rapid collision with the bed jointly contribute to the growth of the granular anisotropy during the initial stage when a is large. Meanwhile, because of the low vertical kinetic energy, the anisotropy is primarily controlled by the gate removal at low a. In addition, the kinetic energy transformation between the vertical and horizontal directions is a major factor influencing the anisotropy of the collapse.

4. Hydroplaning and flow regimes

As the results of the previous section show, the differences in the dynamic processes of submerged particle collapse due to different a and μ_p could lead to a variety of different flow behaviors. In the post-processing section, four dimensionless numbers (Fr_d, N_B, N_S, and N_F) are defined to quantify these different flow behaviors. In this section, we analyze these dimensionless numbers to explore the mechanism of the influence of the particle friction on the dynamic process of submerged particle collapse.

4.1. Hydroplaning

The densimetric Froude number Fr_d is used to quantify the hydroplaning of the submerged particle collapse. The variation in Fr_d with the particle friction coefficient μ_p for different aspect ratios a is shown in Fig. 14. Fr_d decreases as μ_p increases and increases as a increases; therefore, the hydroplaning of particles decreases as μ_p increases. For $a = 0.8$, Fr_d is always less than the lower threshold value of hydroplaning,

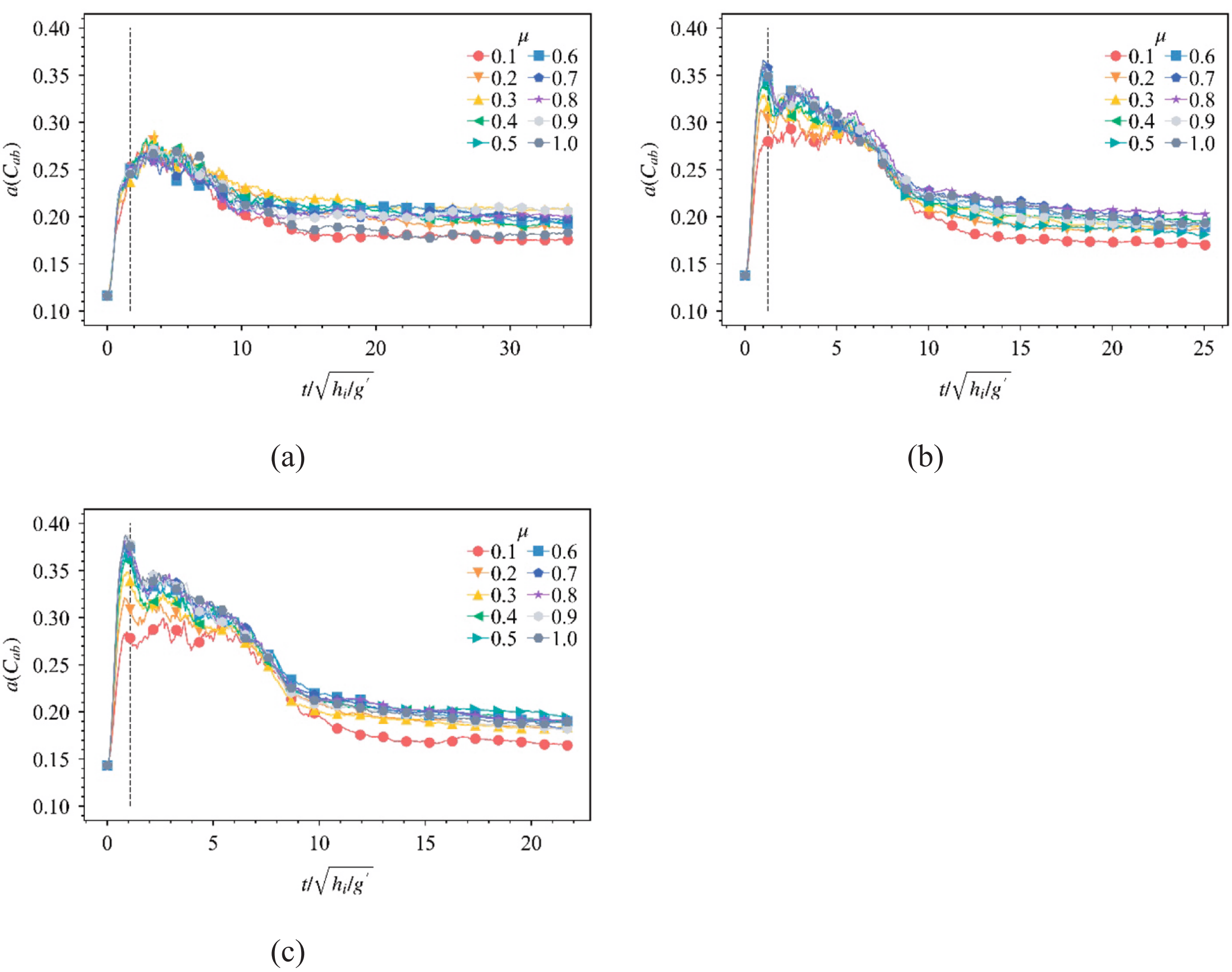


Fig. 13. Comparison of the second invariant of the contact tensor in different cases: **(a)** $a = 0.8$; **(b)** $a = 1.5$; and **(c)** $a = 2$.

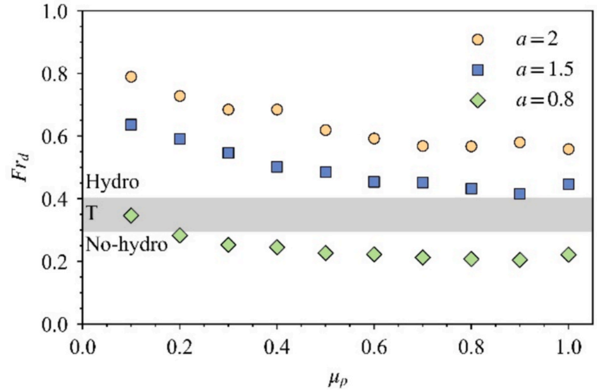


Fig. 14. Relationship between the particle friction coefficient μ_p and the densimetric Froude number Fr_d. Hydro indicates the hydroplaning regime, T indicates the transition regime, and No-hydro indicates the non-hydroplaning regime.

0.4, indicating that the hydroplaning effect is weak in the dynamic process of particle collapse and that the effects of ambient water on the flowing process are not significant. Conversely, the particles are located in the hydroplaning region for $a = 1.5$ and 2.0, which means that the effect of hydroplaning on the flowing behavior is significant. The above results are consistent with the frontal head of the particle flow being uplifted for $\mu_p = 0.4$ and $a = 1.5$ and 2.0, while this phenomenon is not observed at $a = 0.8$ (Fig. 4). According to previous studies (Du et al., 2023; Ilstad et al., 2004a; Mohrig et al., 1998, 1999), when submarine landslides experience hydroplaning, the dynamic pressure at the frontal head uplifts the sliding body, resulting in a water layer being injected into the space between the landslide and the basal bed. This injected water may lubricate the head of the submarine landslide, resulting in a reduction in the basal resistance and thus an increase in the mobility of submarine landslides. Therefore, the relationship between hydroplaning and μ_p and a indicates that the mobility of the particles increases as a increases, but decreases as μ_p increases. Comparing the $\mu_p - Fr_d$ and $\mu_p - L_f^*$ relationships (Fig. 7a), the trend of the variation of L_f^* with μ_p is similar to that of the variation of Fr_d with μ_p, which supports the above-mentioned result concerning the variation in the landslide mobility. Meanwhile, the similar trends of $\mu_p - Fr_d$ and $\mu_p - L_f^*$ also indicate that an inherent correlation exists between Fr_d and L_f^*.

4.2. Flow regimes

The Bagnold number N_B, Savage number N_S, and Friction number N_F are used to characterize the stresses (particle collisional, frictional, and fluid viscous stresses) that dominate the flow.

The Bagnold number N_B and the Savage number N_S vary with μ_p and a, as shown in Fig. 15. N_B and N_S both increase as a increases and decrease as μ_p increases. As previously mentioned, N_B and N_S represent the relative importance of collisional stresses to the viscous stresses and of collisional stresses to the frictional stresses, respectively (Savage 1984). Therefore, the varying trends of N_B and N_S with a indicate that, as a increases, the role of collisional resistance in the flow tends to increase. This result is consistent with the previous analysis of the kinetic energy and anisotropy, which showed that, as a increases, the collapse process of the particles is characterized by higher velocities and the collisions between particles increase.

The relationships of N_B and N_S vary with μ_p and indicate that, as μ_p increases, the proportion of collisional resistance in the flow decreases while both the viscous resistance and the frictional resistance increase. This result is consistent with the general understanding that increasing the particle friction coefficient will result in an increase in the frictional force in the granular flow (Lai et al. 2023). It is also consistent with the observation that the mobility of the particles and the collisions in the flow are reduced when μ_p increases. Notably, the slopes of both the $\mu_p - N_B$ and $\mu_p - N_S$ curves decrease when μ_p is greater than 0.6, which indicates that the resistance force in the flow tends to be similar at high μ_p. This result explains why the influence of μ_p on the runout distance and kinetic energy of the particles tends to be limited when μ_p is high (Sections 3.2 and 3.3). Meanwhile, although N_B exhibits significant variations with changes in a and μ_p, its value remains below the critical value of 200, indicating that the flow behavior of the particles is still controlled by fluid viscous resistance. This result is related to the fact that the viscosity used in the simulation leads to the flow state of the particles being located in the weak viscous regime (Section 2.5). Meanwhile, in some cases with lower values of μ_p (0.1 and 0.2), the value of N_S is greater than the critical value of 0.1, indicating that the flow is dominated by particle collision rather than friction. According to the studies of Du et al. (2022) and Choi and Yu (2023), neither experimental nor real-scale submarine landslides are controlled by both the fluid viscosity ($N_B < 200$) and particle collisions ($N_S > 0.1$). Therefore, the low values of μ_p in this study represent only part of the discussion concerning the influence of the particle friction coefficient on submarine landslides, rather than indicating that actual submarine landslides also have these characteristics.

The $\mu_p - Fr_d$ and $\mu_p - N_B$ relationships are similar. Therefore, to

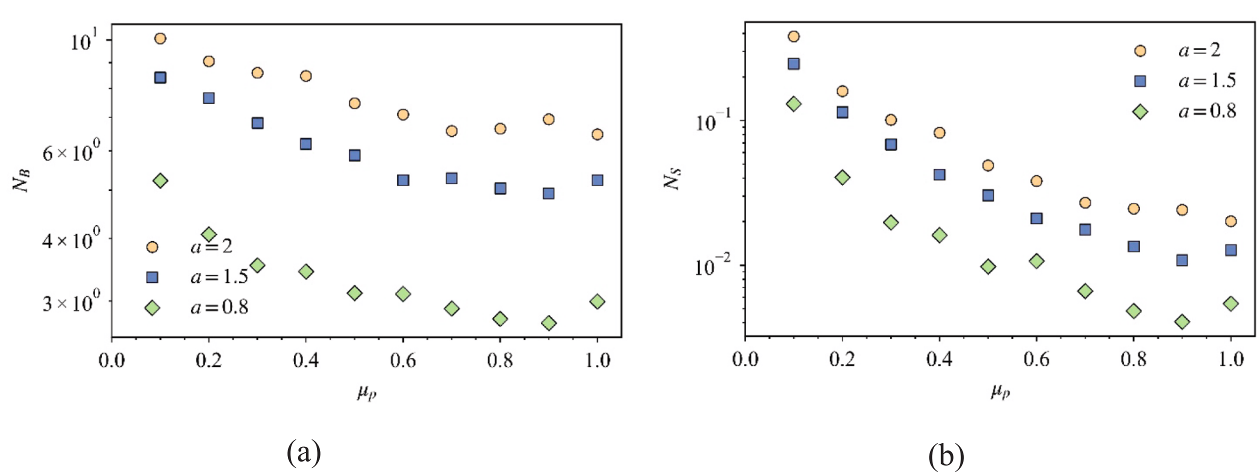


Fig. 15. (a) Bagnold number N_B varying with the particle friction μ_p. **(b)** Savage number N_S varying with the particle friction μ_p.

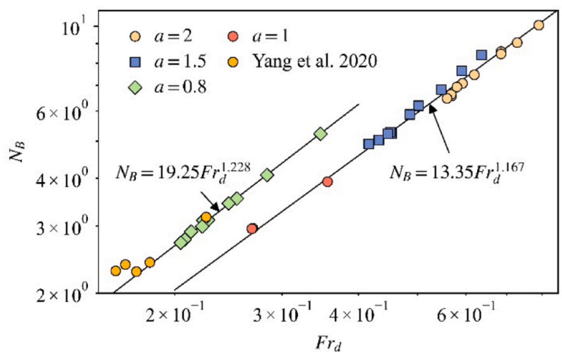


Fig. 16. Relationship between the densimetric Froude number $\mathrm{Fr_d}$ and the Bagnold number $\mathrm{N_B}$. The results of Yang et al. (2020) with a = 0.8 and $\phi_0 \approx$ 0.58 are also plotted. Simulation results with initial conditions of a = 1 and ϕ_0 $\approx$ 0.55, 0.58, and 0.61 are shown.

investigate the inherent relationship between Fr_d and N_B, the Fr_d and N_B values obtained here were analyzed. The variation of N_B with Fr_d for different values of a is shown in Fig. 16. Different relationships exist between Fr_d and N_B for different values of a. At $a = 0.8$, the relationship between Fr_d and N_B is

$$N_B = 19.25Fr_d^{1.228} \tag{29}$$

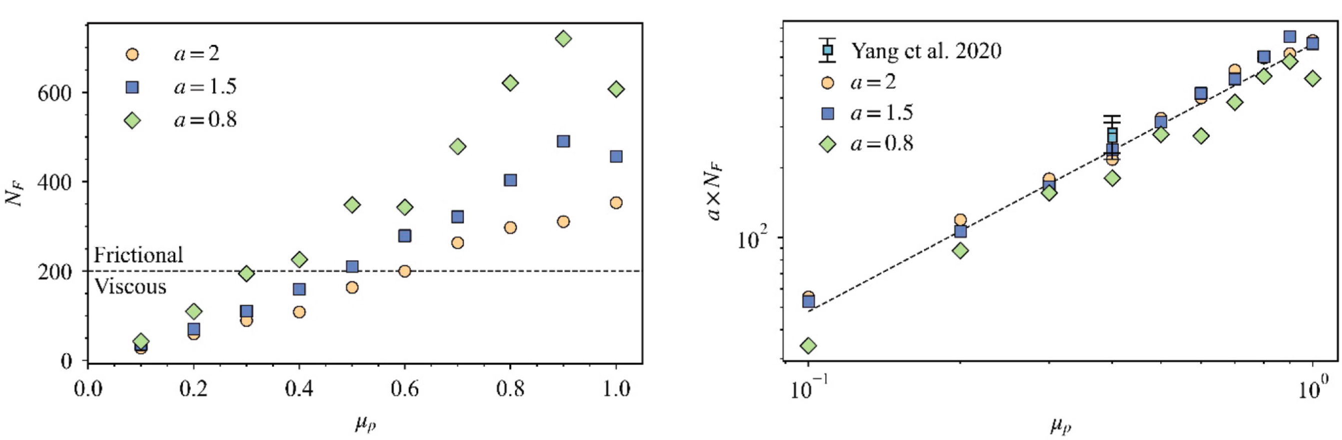
When $a > 0.8$, Fr_d and N_B show a relationship of

$$N_B = 13.35Fr_d^{1.167} \tag{30}$$

To examine the $Fr_d - N_B$ relationship given in this study, the Fr_d and N_B results obtained from simulations with $a = 1$, $\mu_p = 0.4$, and $\phi_0 \approx 0.55$, 0.58, and 0.61 (with other simulation conditions remaining the same) and that obtained by Yang et al. (2020) with $a = 0.8$, $\mu_p = 0.4$, and $\phi_0 \approx$ 0.58 are plotted in Fig. 16. It can be seen that, whether $a = 0.8$ or 1.0, the values of Fr_d and N_B are consistently located near the fitting curve given in this study. The Fr_d and N_B values obtained by Yang et al. (2020) fall near the fitting curve, indicating that the $Fr_d - N_B$ relationship given in this study is robust. Moreover, the Fr_d and N_B values obtained at different ϕ_0 also fall close to the fitting curve, indicating that the $Fr_d - N_B$ relationship given in this study is independent of the value of ϕ_0. In fact, ignoring the constant value in N_B (Eq. (16)), a relationship of $N_B \sim \phi_{ft}v_{ft}/((1-\phi_{ft})h_{ft})$ can be obtained. For Fr_d, we also have $Fr_d \sim v_{ft}/\sqrt{h_{ft}}$. Therefore, Fr_d and N_B are theoretically related and their relationship can be simplified as $N_B \sim \phi_{ft}Fr_d/((1-\phi_{ft})\sqrt{h_{ft}})$. The difference in the relationship between Fr_d and N_B at $a = 0.8$ compared with at $a > 0.8$ suggests that there is a different collapse pattern (as Section 3.1 shows), which leads to a difference in the evolutions of ϕ_{ft} and h_{ft} in the flow when $a = 0.8$. Conversely, when $a > 0.8$, the evolutions of ϕ_{ft} and h_{ft} at the front of the flow tend to be similar.

The Friction number N_F represents the relative importance of the frictional resistance and the fluid viscous resistance in submarine landslides; therefore, using N_F to quantify the influence of the particle friction coefficient on the flow is very important to understanding the flow behavior of submarine landslides. The N_F of the flows is calculated using Eq. (19), and the variation of N_F with μ_p is shown in Fig. 17. N_F is seen to increase as μ_p increases. Meanwhile, when μ_p is small, because of the lower frontal head depth h_{ft} and higher frontal velocity v_{ft}, N_F is lower or close to the critical value of 200, indicating that the granular flow is dominated by the fluid viscous resistance under this condition. Conversely, when μ_p is large, N_F is higher than the critical value of 200, indicating that h_{ft} and v_{ft} in the flow are increasing and decreasing, respectively. This indicates that the flow is dominated by frictional resistance when μ_p is large, which is consistent with the analysis of the flow state (Fig. 4) and N_S (Fig. 15b) in the previous section. We also observe that N_F decreases as a increases in Fig. 17a, which indicates that the corresponding value of μ_p is different when N_F is at the critical value of 200 and a differs. The $a - N_F$ relationship is related to the fact that the flow velocity v_{ft} increases as a increases, while the frontal head depth h_{ft} tends to a certain value as a increases.

As previously mentioned, N_F represents the relative importance of the frictional resistance to the fluid viscous resistance. Therefore, we conducted a series of dry particle collapse simulations (no fluid effect) to comparatively explore the influence of μ_p on the flow dynamics. The normalized runout distance L_f* of the dry particle collapse is plotted in Fig. 7. The pattern of the runout distance of the dry particles varies with μ_p and a is similar to that of the submerged particles. However, when a = 1.5and 2.0 and the particles have a small value of μ_p, the L_f* value of the submerged particles is higher than that of the dry particles, indicating that submerged particles have higher mobility. Comparing Fig. 7 and Fig. 17a, it can be seen that, in the case where the L_f* value of the submerged particle is higher, the flow states basically correspond to a state in which the particles are dominated by the fluid viscous stress. To make this statement clearer, we used the threshold value $N_F = 200$ and classified all the cases into two categories: those controlled by fluid viscous stress ($N_F \leq 200$, viscous) and those controlled by particle frictional stress ($N_F > 200$, frictional). After that, we separate the two



(a) (b)

Fig. 17. **(a)** Friction number N_F varying with the particle friction coefficient μ_p. **(b)** Relationship between the Friction number N_F and the particle friction coefficient μ_p after N_F is scaled by the initial aspect ratio a.

categories with a dotted line in Fig. 7a. It can be more clearly seen that, when the flow is dominated by the fluid viscous stress, the L_f* value of the submerged particles is greater than that of the dry particles, while when the flow is dominated by frictional stress, the difference in L_f* between the submerged particles and the dry particles is not significant. This is because, when the flow is dominated by the fluid viscous stress, the effects of eddies generated by the particle collapse and the lubrication effects of the ambient water may be stronger (Section 3.1), leading to an increase in L_f* for submerged particles. One interesting phenomenon worth noting is that the L_f* of the cases where $a = 0.8$ are similar for dry and submerged conditions, no matter whether the flow is dominated by fluid viscous stress or particle frictional stress. This phenomenon could be explained by the influence of the hydroplaning effect. For the condition where $a = 0.8$, although when $\mu_p \leq 0.3$, the system is controlled by fluid viscous stress ($N_F \leq 200$), the overall particle collapse pattern has changed (Section 3.1 and 3.2), resulting in a fact that the flowing particles didn't generate hydroplaning (Section 4.1, when $a =$ 0.8, the Fr_d value is less than 0.4). Previous studies have shown that hydroplaning can prevent the transfer of resistance between the flowing landslide body and the base, which is the main reason for the flow resistance reduction and long runout distance of submarine landslides. Therefore, when there is no hydroplaning effect in the system, compared to the cases with $\mu_p > 0.3$, although the system is relatively more strongly influenced by eddies generated by the particle collapse and the lubrication effects of the ambient water when μ_p is less than 0.3, the system is still dominated by no-hydroplaning, resulting in an insignificant reduction in overall flow resistance. Therefore, ultimately, we observe a similar runout distance for dry and submerged conditions. This result once again indicates that the hydroplaning effect is a dominant factor that controls the runout distance of the submarine landslides.

As previously mentioned, N_F is dependent on a; therefore, the aspect ratio a is used as a variable to scale N_F further to explore the relationship between N_F, a, and μ_p. The scaling results are shown in Fig. 17b. There is a good linear relationship between $a \times N_F$ and μ_p in the double logarithmic coordinate system, indicating that $a \times N_F$ non-linearly increases as μ_p increases. To examine the $a \times N_F - \mu_p$ relationship given in this study, the results of Yang et al. (2020) with $\mu_p = 0.4$ and $a = 0.8$ are also plotted. It can be seen that the values of a, N_F, and μ_p given by Yang et al. (2020) fall near the fitted curve of this study, confirming the robustness of the fitting results. The significance of the relationship between $a \times N_F$ and μ_p is that, in submarine landslides or submerged particle collapse, if the initial values of a and μ_p are given, it is possible to evaluate N_F and to estimate whether the flow is controlled by frictional or fluid viscous resistance. Therefore, via comparisons with sub-aerial landslides, it is possible to roughly estimate the runout distance of submarine landslides. As Fig. 7 shows, the runout distances of submarine landslides are greater than those of subaerial landslides when the landslide is controlled by fluid viscous resistance. Meanwhile, when the flow is controlled by frictional resistance, the difference in the runout distances between subaerial landslides and submarine landslides is not significant.

5. Discussion

5.1. Runout distance scaling

In the analysis of the densimetric Froude number Fr_d, it was found that the normalized runout distance of submerged particle collapse L_f* has an inherent correlation with Fr_d. We conducted a more detailed analysis of Fr_d and L_f* to explore this inherent relationship and to build an empirical relationship to estimate the runout distances of submarine landslides. The fitting result of Fr_d and L_f* is shown in Fig. 18. In the range of Fr_d from 0.1 to 0.8, Fr_d and L_f* have a positive linear correlation in the semi-logarithmic coordinate system, which can be expressed as

$$L_f^* = 10^{1.308Fr_d - 0.204} \tag{31}$$

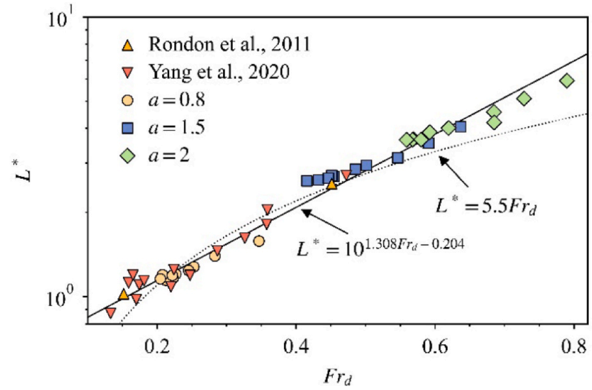


Fig. 18. Relationship between the densimetric Froude number Fr_d and the normalized runout distance L_f*. The results of Rondon et al. (2011) and Yang et al. (2020) are also plotted. The dotted line shows the $Fr_d - L_f^*$ relationship proposed by Yang et al. (2020).

Different values of a and μ_p were used to obtain the $Fr_d - L_f^*$ relationship; therefore, the relationship between Fr_d and L_f* presented in this study is independent of a and μ_p. The coefficient of determination (R^2) is as high as 0.98. To verify the $Fr_d - L_f^*$ relationship given in this study, the experimental results of Rondon et al. (2011) and the simulation results of Yang et al. (2020) are also plotted. Note that the Fr_d value of Rondon et al. (2011) is calculated according to the procedures of Yang et al. (2020). It can be seen that the results of these previous studies are all located near the fitting curve, indicating the validity and robustness of the $Fr_d - L_f^*$ relationship given in this study.

The studies of Yang et al. (2020) and Sun et al. (2020) also provided a $Fr_d - L_f^*$ relationship, i.e., $L_f^* = 5.5 \times Fr_d$. We plotted their relationship in Fig. 18. When the Fr_d value is within the range of [0.2, 0.5], the curve given by Yang et al. (2020) and Sun et al. (2020) can describe the $Fr_d - L_f^*$ relationship. However, their curve deviates from measured Fr_d and L_f* values when Fr_d is outside this range. Meanwhile, the $Fr_d - L_f^*$ relationship given in this study can describe the relationship between Fr_d and L_f* more accurately within the range of $Fr_d = [0.1, 0.8]$, indicating that the $Fr_d - L_f^*$ relationship given in this study has a broader applicability.

Sequeiros (2012) summarized the Fr_d values obtained from submarine landslide experiments and field observations, showing that the Fr_d values of submarine landslides are approximately within the range of 0.01 to 2. In actual submarine landslides, if the $Fr_d - L_f^*$ relationship given by Yang et al. (2020) and Sun et al. (2020) is used, the maximum value of L_f* is approximately 11, which may not explain the runout distances of hundreds or even thousands of kilometers observed in historical submarine landslides. Conversely, the result that L_f* exponentially increases as Fr_d increases given in this study provides a good correlation with the long runout distances of submarine landslides.

5.2. Limitations and future studies

Compared with residual or alluvial deposits that have not undergone long-distance transportation, most submarine sediments—particularly deep-sea sediments involved in submarine landslides—are relatively rounded or well-rounded loose granular materials (Kleesment 2009; Pokhrel et al. 2024). Therefore, in this study, we focused solely on investigating the influence of inter-particle friction on the dynamic processes of submarine landslides, while excluding rolling resistance. However, current research indicates that some submarine landslide materials possess irregular shapes, which result in significant rolling resistance that cannot be ignored (Iwashita 1998; Tordesillas and Walsh 2002). For such submarine landslides, the applicability of our current conclusions requires further discussion. Consequently, future studies should incorporate physical model experiments or numerical simulations to investigate the contribution of rolling resistance to the dynamic processes of submarine landslides, especially when rolling friction is coupled with inter-particle friction. As rolling resistance is related to particle shape, material deformation at contact points, surface roughness, and other factors (Estrada et al. 2011; Abdelbary and Chang 2023), and since particle shape is considered a key controlling factor, adopting the rolling friction model in numerical simulations or directly studying the influence of particle shape on flow would be promising directions for future research (Luding 2008).

In addition, when using the particle collapse model to study the physical processes of disasters such as landslides and debris flows, the aspect ratio of the particle column has been a key topic of discussion (Topin et al. 2012; Bougouin and Lacaze 2018; Jing et al. 2018; Sun et al. 2021). As mentioned in Section 3.2, it is widely accepted that a critical aspect ratio exists, beyond which significant differences in particle dynamics occur. For example, Topin et al. (2012) found that below the critical aspect ratio a_c, the runout distance of submerged particles increases linearly with the value of a. However, when $a > a_c$, the runout distance of the particles increases exponentially with the increase of a. Similarly, Jing et al. (2018) found that the runout distance of particle increases in a segmented power-law manner as a increases. They attribute the change in the relationship between the normalized runout distance and the initial aspect ratio at a_c to the different growth rates of kinetic energy in the vertical and horizontal directions. Although our results suggest that under the simulation conditions of this study, the a_c might fall within the range of $a = 0.8$ to 1.5 (see Section 3.2 for details on dynamic characteristics), the limitations of the LBM-DEM method prevented us from simulating cases with higher aspect ratios to determine a_c or to study the influence of particle friction on the dynamics at high aspect ratios. Previous studies have indicated that the error in the LBM solution is on the order of Ma^2 when simulating incompressible fluid flow (Ma is Mach number) (Mohamad 2019). After conducting a trial-and-error analysis, we found that when $a \le 2.0$, the simulation conditions satisfied the Mach number requirements. On the contrary, when a > 2, the Mach number of the system does not meet the requirements of incompressible fluid. As a result, the study did not include high aspect ratio cases due to the Mach number condition. It is worth noting that, in our analysis of runout scaling (Section 5.1), we incorporated data obtained from Rondon et al. (2011) under high aspect ratio conditions. Therefore, our conclusion that the flow distance in submarine landslides has a non-linear relationship with *Frd* is applicable even for high aspect ratios. Nevertheless, given the significance of the aspect ratio, we recommend that future research focus on investigating the influence of particle friction on flow under high aspect ratios.

6. Conclusions

In this study, a series of LBM-DEM simulations was conducted to provide micromechanical insights into the crucial role of particle friction on the initiation and dynamic processes of submarine landslides. On the basis of analyses of the flow behavior, runout distance, and microscopic contact, the particle friction and aspect ratio effects were comparatively studied and discussed in detail. The main findings are summarized as follows.

The particle friction can significantly influence the dynamic processes of submarine landslides. For slide initiation, increasing the particle friction will cause the triggering time to increase, resulting in a slower collapse process. In addition, as the particle friction increases, the area of the dead zone will increase, leading to a decrease in the number of moving particles. Because the lubrication and buoyancy of water reduce the friction between particles, the particle column is still sliding-dominated at high particle friction but not toppling-dominated, as found in dry systems.

The influence of the particle friction on the runout distance of the granules is nonlinear, meaning that the runout distance decreases exponentially with increasing particle friction. Compared with a dry system, the runout distance will be higher when the submerged particles are dominated by fluid viscous resistance, while the runout distances under dry and submerged conditions are similar when the particles are dominated by friction.

The mechanism by which particle friction affects the runout distance is that, as the friction between the particles increases, the directions of the strong force chains in the system tend to concentrate upward, leading to a decrease in the number of strong force chains driving the particle movement.

There is a significant correlation between the Bagnold number N_B and the densimetric Froude number Fr_d, which depends on the initial aspect ratio a. For $a = 0.8$, the relationship between N_B and Fr_d is $N_B = 19.25Fr_d^{1.228}$, while it is $N_B = 19.25Fr_d^{1.228}$ when $a > 0.8$. In addition, the Friction number N_F is highly correlated with the particle friction μ_p and, when N_F is scaled by the aspect ratio a, a relationship between a, N_F, and μ_p can be established.

The densimetric Froude number Fr_d and the normalized runout distance L_f* have an inherent relationship of $L_f^* = 10^{1.308Fr_d - 0.204}$. The result in which L_f* exponentially increases as Fr_d increases given in this study is in good agreement with the long runout distances of submarine landslides.

CRediT authorship contribution statement

Shu Zhou: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yandong Bi:** Writing – review & editing, Visualization, Methodology, Formal analysis, Conceptualization. **Xiaolin Tan:** Writing – review & editing, Investigation, Formal analysis. **Zhen Guo:** Writing – review & editing, Validation, Supervision. **Chongqiang Zhu:** Writing – review & editing, Supervision, Software. **Jin Sun:** Writing – review & editing, Supervision, Software. **Yu Huang:** Writing – review & editing, Supervision, Software, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors acknowledge financial support from the National Natural Science Foundation of China (Grant No. 42120104008) and (Grant No. 42507271).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compgeo.2026.108049.

Data availability

Data will be made available on request.

References

Abdelbary A, Chang L (2023) Principles of Engineering Tribology: Fundamentals and Applications. Elsevier.

Bagnold, R.A., 1954. Experiments on a gravity-free dispersion of large solid spheres in a Newtonian fluid under shear. Proc. R. Soc. Lond. A 225, 49–63. https://doi.org/10.1098/rspa.1954.0186.

Bhatnagar, P.L., Gross, E.P., Krook, M., 1954. A model for collision processes in gases. i. small amplitude processes in charged and neutral one-component systems. Phys. Rev. 94, 511–525. https://doi.org/10.1103/PhysRev.94.511.

Blott, S.J., Pye, K., 2008. Particle shape: a review and new methods of characterization and classification. Sedimentology 55, 31–63. https://doi.org/10.1111/j.1365-3091.2007.00892.x.

Bougouin, A., Lacaze, L., 2018. Granular collapse in a fluid: different flow regimes for an initially dense-packing. Phys. Rev. Fluids 3, 064305. https://doi.org/10.1103/PhysRevFluids.3.064305.

Bougouin, A., Lacaze, L., Bonometti, T., 2017. Collapse of a neutrally buoyant suspension column: from Newtonian to apparent non-Newtonian flow regimes. J. Fluid Mech. 826, 918–941. https://doi.org/10.1017/jfm.2017.471.

Brown, P.P., Lawler, D.F., 2003. Sphere Drag and Settling Velocity Revisited. J. Environ. Eng. 129, 222–231. https://doi.org/10.1061/(ASCE)0733-9372(2003)129:3(222).

Choi, C.E., Yu, J., 2023. Mesoscopic dimensional analysis of submarine debris flows: implications for physical modelling. Granul. Matter 25, 14. https://doi.org/10.1007/s10035-022-01294-8.

Cundall, P.A., Strack, O.D.L., 1979. A discrete numerical model for granular assemblies. Géotechnique 29, 47–65. https://doi.org/10.1680/geot.1979.29.1.47.

De Blasio, F.V., Engvik, L., Harbitz, C.B., Elverhøi, A., 2004. Hydroplaning and submarine debris flows. J. Geophys. Res. Oceans 109. https://doi.org/10.1029/2002JC001714.

Deng, J., Zhang, X., Shen, S., Koseki, J., 2018. Low friction coefficient (approximately tan1°) of subaqueous debris flow in rotating flume tests and its mechanism. Bull. Eng. Geol. Env. 77, 931–939. https://doi.org/10.1007/s10064-017-1081-9.

Du, J., Choi, C.E., Yu, J., Thakur, V., 2022. Mechanisms of Submarine Debris Flow Growth. J. Geophys. Res. Earth 127, e2021JF006470. https://doi.org/10.1029/2021JF006470.

Du, J., Yu, J., Choi, C.E., 2023. Effects of bed permeability and roughness on the mobility of submarine debris flows: experimental insights. Landslides 20, 497–510. https://doi.org/10.1007/s10346-022-02005-3.

Elverhoi, A., Breien, H., De Blasio, F.V., et al., 2010. Submarine landslides and the importance of the initial sediment composition for run-out length and final deposit. Ocean Dyn. 60, 1027–1046. https://doi.org/10.1007/s10236-010-0317-z.

Estrada, N., Azéma, E., Radjai, F., Taboada, A., 2011. Identification of rolling resistance as a shape parameter in sheared granular media. Phys. Rev. E 84, 011306. https://doi.org/10.1103/PhysRevE.84.011306.

Guo Z, Shu C (2013) Lattice Boltzmann Method And Its Application In Engineering, Illustrated edition. World Scientific.

Guo, X., Zheng, D., Nian, T., Yin, P., 2019. Effect of different span heights on the pipeline impact forces induced by deep-sea landslides. Appl. Ocean Res. 87, 38–46. https://doi.org/10.1016/j.apor.2019.03.009.

Hampton, M.A., Lee, H.J., Locat, J., 1996. Submarine landslides. Rev. Geophys. 34, 33–59. https://doi.org/10.1029/95RG03287.

He, X., Luo, L.-S., 1997. Lattice Boltzmann Model for the Incompressible Navier–Stokes Equation. J. Stat. Phys. 88, 927–944. https://doi.org/10.1023/B:JOSS.0000015179.12689.e4.

Hsu, S.-K., Kuo, J., Chung-Liang, L., et al., 2008. Turbidity currents, submarine landslides and the 2006 Pingtung earthquake off SW Taiwan. Terr. Atmos. Ocean. Sci. 19, 7. https://doi.org/10.3319/TAO.2008.19.6.767(PT).

Ilstad, T., De Blasio, F.V., Elverhøi, A., et al., 2004a. On the frontal dynamics and morphology of submarine debris flows. Mar. Geol. 213, 481–497. https://doi.org/10.1016/j.margeo.2004.10.020.

Ilstad, T., Elverhøi, A., Issler, D., Marr, J.G., 2004b. Subaqueous debris flow behaviour and its dependence on the sand/clay ratio: a laboratory study using particle tracking. Mar. Geol. 213, 415–438. https://doi.org/10.1016/j.margeo.2004.10.017.

Iverson, R.M., 1997. The physics of debris flows. Rev. Geophys. 35, 245–296. https://doi.org/10.1029/97RG00426.

Iwashita, O.a., 1998. Rolling resistance at contacts in simulation of shear band development by DEM. J. Eng. Mech. 124.

Izard, E., Lacaze, L., Bonometti, T., Pedrono, A., 2018. Numerical Modeling of a Granular Collapse Immersed in a Viscous Fluid. In: Gourbesville, P., Cunge, J., Caignaert, G. (Eds.), Advances in Hydroinformatics. Springer, Singapore, pp. 1099–1116.

Jing, L., Yang, G.C., Kwok, C.Y., Sobral, Y.D., 2018. Dynamics and scaling laws of underwater granular collapse with varying aspect ratios. Phys. Rev. E 98, 042901. https://doi.org/10.1103/PhysRevE.98.042901.

Jing, L., Yang, G.C., Kwok, C.Y., Sobral, Y.D., 2019. Flow regimes and dynamic similarity of immersed granular collapse: a CFD-DEM investigation. Powder Technol. 345, 532–543. https://doi.org/10.1016/j.powtec.2019.01.029.

Karstens, J., Haflidason, H., Berndt, C., Crutchley, G.J., 2023. Revised Storegga Slide reconstruction reveals two major submarine landslides 12,000 years apart. Commun. Earth Environ. 4, 1–9. https://doi.org/10.1038/s43247-023-00710-y.

Kleesment, A., 2009. Roundness and surface features of quartz grains in Middle Devonian deposits of the East Baltic and their palaeogeographical implications. Estonian J. Earth Sci. 58. https://doi.org/10.3176/earth.2009.1.07.

Krüger T, Kusumaatmaja H, Kuzmin A, et al (2016) The Lattice Boltzmann Method: Principles and Practice, 1st ed. 2017 edition. Springer, New York, NY.

Kumar, K., Delenne, J.-Y., Soga, K., 2017. Mechanics of granular column collapse in fluid at varying slope angles. J. Hydrodyn. Ser B 29, 529–541. https://doi.org/10.1016/S1001-6058(16)60766-7.

Lacaze, L., Bouteloup, J., Fry, B., Izard, E., 2021. Immersed granular collapse: from viscous to free-fall unsteady granular flows. J. Fluid Mech. 912, A15. https://doi.org/10.1017/jfm.2020.1088.

Lai, Z., Jiang, E., Zhao, L., et al., 2023. Granular column collapse: Analysis of inter-particle friction effects. Powder Technol. 415, 118171. https://doi.org/10.1016/j.powtec.2022.118171.

Lajeunesse, E., Monnier, J.B., Homsy, G.M., 2005. Granular slumping on a horizontal surface. Phys. Fluids 17, 103302. https://doi.org/10.1063/1.2087687.

Li, Y., Xu, Y., Thornton, C., 2005. A comparison of discrete element simulations and experiments for 'sandpiles' composed of spherical particles. Powder Technol. 160, 219–228. https://doi.org/10.1016/j.powtec.2005.09.002.

Li, D., Li, Y., Wang, Z., et al., 2016. Quantitative, SEM-based shape analysis of sediment particles in the Yellow River. Int. J. Sedim. Res. 31, 341–350. https://doi.org/10.1016/j.ijsrc.2016.05.006.

Li, L., Shi, F., Ma, G., Qiu, Q., 2019. Tsunamigenic potential of the Baiyun Slide complex in the South China Sea. J. Geophys. Res. Solid Earth 124, 7680–7698. https://doi.org/10.1029/2019JB018062.

Liu, D., Cui, Y., Guo, J., et al., 2020. Investigating the effects of clay/sand content on depositional mechanisms of submarine debris flows through physical and numerical modeling. Landslides 17, 1863–1880. https://doi.org/10.1007/s10346-020-01387-6.

Locat, J., Lee, H.J., 2002. Submarine landslides: advances and challenges. Can. Geotech. J. 39, 193–212. https://doi.org/10.1139/t01-089.

Lootens, D., van Damme, H., Hémar, Y., Hébraud, P., 2005. Dilatant flow of concentrated suspensions of rough particles. Phys. Rev. Lett. 95, 268302. https://doi.org/10.1103/PhysRevLett.95.268302.

Luding, S., 2008. Cohesive, frictional powders: contact models for tension. Granul. Matter 10, 235–246. https://doi.org/10.1007/s10035-008-0099-x.

Lu, Y., Liu, X., Xie, X., et al., 2024. Particle-scale analysis on dynamic response of turbidity currents to sediment concentration and bedforms. Phys. Fluids 36, 033316. https://doi.org/10.1063/5.0191219.

Marr, J.G., Harff, P.A., Shanmugam, G., Parker, G., 2001. Experiments on subaqueous sandy gravity flows: the role of clay and water content in flow dynamics and depositional structures. Geol. Soc. Am. Bull. 113, 1377–1386. https://doi.org/10.1130/0016-7606(2001)113%253C1377:EOSSGF%253E2.0.CO;2.

Masson, D.G., Harbitz, C.B., Wynn, R.B., et al., 2006. Submarine landslides: processes, triggers and hazard prediction. Philos. Trans. R. Soc. A Math. Phys. Eng. Sci. 364, 2009–2039. https://doi.org/10.1098/rsta.2006.1810.

Meruane, C., Tamburrino, A., Roche, O., 2010. On the role of the ambient fluid on gravitational granular flow dynamics. J. Fluid Mech. 648, 381–404. https://doi.org/10.1017/S0022112009993181.

MiDi, G.D.R., 2004. On dense granular flows. Eur. Phys. J. E 14, 341–365. https://doi.org/10.1140/epje/i2003-10153-0.

Mindlin, R.D., 1949. Compliance of elastic bodies in contact. J. Appl. Mech. 16, 259–268. https://doi.org/10.1115/1.4009973.

Mohamad AA (2019) Lattice Boltzmann Method: Fundamentals and Engineering Applications with Computer Codes. Springer, London.

Mohrig, D., Marr, J.G., 2003. Constraining the efficiency of turbidity current generation from submarine debris flows and slides using laboratory experiments. Mar. Pet. Geol. 20, 883–899. https://doi.org/10.1016/j.marpetgeo.2003.03.002.

Mohrig, D., Ellis, C., Parker, G., et al., 1998. Hydroplaning of subaqueous debris flows. GSA Bull. 110, 387–394. https://doi.org/10.1130/0016-7606(1998)110%253C0387:HOSDF%253E2.3.CO;2.

Mohrig, D., Elverhøi, A., Parker, G., 1999. Experiments on the relative mobility of muddy subaqueous and subaerial debris flows, and their capacity to remobilize antecedent deposits. Mar. Geol. 154, 117–129. https://doi.org/10.1016/S0025-3227(98)00107-8.

Moon, J.Y., Dai, S., Chang, L., et al., 2015. The effect of sphere roughness on the rheology of concentrated suspensions. J. Nonnewton. Fluid Mech. 223, 233–239. https://doi.org/10.1016/j.jnnfm.2015.07.007.

Mosher, D.C., Moscardelli, L., Shipp, R.C., et al., 2010. Submarine Mass Movements and their Consequences. In: Mosher, D.C., Shipp, R.C., Moscardelli, L. (Eds.), Submarine Mass Movements and Their Consequences. Springer, Netherlands, Dordrecht, pp. 1–8.

Najuch, T., Sun, J., 2019. Analysis of two partially-saturated-cell methods for lattice Boltzmann simulation of granular suspension rheology. Comput. Fluids 189, 1–12. https://doi.org/10.1016/j.compfluid.2019.05.004.

Ness, C., Sun, J., 2015. Flow regime transitions in dense non-Brownian suspensions: rheology, microstructural characterization, and constitutive modeling. Phys. Rev. E Stat. Nonlin. Soft Matter Phys. 91, 012201. https://doi.org/10.1103/PhysRevE.91.012201.

Ng, C.W.W., Choi, C.E., Law, R.P.H., 2013. Longitudinal spreading of granular flow in trapezoidal channels. Geomorphology 194, 84–93. https://doi.org/10.1016/j.geomorph.2013.04.016.

Ng, C.W.W., Choi, C.E., Goodwin, G.R., 2019. Froude characterization for unsteady single-surge dry granular flows: Impact pressure and runup height. Can. Geotech. J. 56, 1968–1978. https://doi.org/10.1139/cgj-2018-0529.

Noble, D.R., Torczynski, J.R., 1998. A Lattice-Boltzmann method for partially saturated computational cells. Int. J. Mod. Phys. C 09, 1189–1201. https://doi.org/10.1142/S0129183198001084.

Owen, D.R.J., Leonardi, C.R., Feng, Y.T., 2011. An efficient framework for fluid–structure interaction using the lattice Boltzmann method and immersed moving boundaries. Int. J. Numer. Meth. Eng. 87, 66–95. https://doi.org/10.1002/nme.2985.

Parsons, J., D.,, et al., 2001. Experimental Study of the Grain-Flow, Fluid-Mud transition in Debris Flows. J. Geol.

Pokhrel, P., Attal, M., Sinclair, H.D., et al., 2024. Downstream rounding rate of pebbles in the Himalaya. Earth Surf. Dyn. 12, 515–536. https://doi.org/10.5194/esurf-12-515-2024.

Ren, Z., Zhao, X., Liu, H., 2019. Numerical study of the landslide tsunami in the South China Sea using Herschel-Bulkley rheological theory. Phys. Fluids 31, 056601. https://doi.org/10.1063/1.5087245.

Rojek, J., 2018. Contact Modeling in the Discrete Element Method. In: Popp, A., Wriggers, P. (Eds.), Contact Modeling for Solids and Particles. Springer International Publishing, Cham, pp. 177–228.

Rondon, L., Pouliquen, O., Aussillous, P., 2011. Granular collapse in a fluid: Role of the initial volume fraction. Phys. Fluids 23, 073301. https://doi.org/10.1063/1.3594200.

Savage, S.B., 1984. The Mechanics of Rapid Granular Flows. In: Hutchinson, J.W., Wu, T. Y. (Eds.), Advances in Applied Mechanics. Elsevier, pp. 289–366.

Sawyer, D.E., Flemings, P.B., Buttles, J., Mohrig, D., 2012. Mudflow transport behavior and deposit morphology: role of shear stress to yield strength ratio in subaqueous experiments. Mar. Geol. 307–310, 28–39. https://doi.org/10.1016/j.margeo.2012.01.009.

Schnyder, J.S., Eberli, G.P., Kirby, J.T., et al., 2016. Tsunamis caused by submarine slope failures along western Great Bahama Bank. Sci. Rep. 6, 35925. https://doi.org/10.1038/srep35925.

Schulten, I., Mosher, D.C., Piper, D.J.W., Krastel, S., 2019. A massive slump on the St. Pierre Slope, a new perspective on the 1929 grand banks submarine landslide. J. Geophys. Res. Solid Earth 124, 7538–7561. https://doi.org/10.1029/2018JB017066.

Seiphoori, A., Gunn, A., Kosgodagan Acharige, S., et al., 2021. Tuning sedimentation through surface charge and particle shape. Geophys. Res. Lett. 48, e2020GL091251. https://doi.org/10.1029/2020GL091251.

Sequeiros, O.E., 2012. Estimating turbidity current conditions from channel morphology: a Froude number approach. J. Geophys. Res. Oceans 117. https://doi.org/10.1029/2011JC007201.

Shen, Z., Wang, G., Huang, D., Jin, F., 2022. A resolved CFD-DEM coupling model for modeling two-phase fluids interaction with irregularly shaped particles. J. Comput. Phys. 448, 110695. https://doi.org/10.1016/j.jcp.2021.110695.

Sivadasan, V., Lorenz, E., Hoekstra, A.G., Bonn, D., 2019. Shear thickening of dense suspensions: the role of friction. Phys. Fluids 31, 103103. https://doi.org/10.1063/1.5121536.

Song, P., Choi, C.E., 2021. Revealing the Importance of Capillary and Collisional Stresses on Soil Bed erosion Induced by Debris Flows. J. Geophys. Res. Earth 126, e2020JF005930. https://doi.org/10.1029/2020JF005930.

Song, D., Choi, C.E., Ng, C.W.W., Zhou, G.G.D., 2018. Geophysical flows impacting a flexible barrier: Effects of solid-fluid interaction. Landslides 15, 99–110. https://doi.org/10.1007/s10346-017-0856-1.

Staron, L., Hinch, E.J., 2005. Study of the collapse of granular columns using two-dimensional discrete-grain simulation. J. Fluid Mech. 545, 1–27. https://doi.org/10.1017/S0022112005006415.

Stevenson, C.J., Feldens, P., Georgiopoulou, A., et al., 2018. Reconstructing the sediment concentration of a giant submarine gravity flow. Nat. Commun. 9, 2616. https://doi.org/10.1038/s41467-018-05042-6.

Succi S (2001) The Lattice Boltzmann Equation for Fluid Dynamics and Beyond, 1st edition. Clarendon Press, Oxford : New York.

Sun, Y., Zhang, W., Wang, X., Liu, Q., 2020. Numerical study on immersed granular collapse in viscous regime by particle-scale simulation. Phys. Fluids 32, 073313. https://doi.org/10.1063/5.0015110.

Sun, Y., Zhang, W., An, Y., et al., 2021. Experimental investigation of immersed granular collapse in viscous and inertial regimes. Phys. Fluids 33, 103317. https://doi.org/10.1063/5.0067485.

Talling, P.J., Wynn, R.B., Masson, D.G., et al., 2007. Onset of submarine debris flow deposition far from original giant landslide. Nature 450, 541–544. https://doi.org/10.1038/nature06313.

Tappin, D.R., Grilli, S.T., Harris, J.C., et al., 2014. Did a submarine landslide contribute to the 2011 Tohoku tsunami? Mar. Geol. 357, 344–361. https://doi.org/10.1016/j.margeo.2014.09.043.

Topin, V., Monerie, Y., Perales, F., Radjaï, F., 2012. Collapse Dynamics and Runout of Dense Granular Materials in a Fluid. Phys. Rev. Lett. 109, 188001. https://doi.org/10.1103/PhysRevLett.109.188001.

Tordesillas, A., Walsh, D.C.S., 2002. Incorporating rolling resistance and contact anisotropy in micromechanical models of granular media. Powder Technol. 124, 106–111. https://doi.org/10.1016/S0032-5910(01)00490-9.

Urlaub, M., Talling, P.J., Zervos, A., Masson, D., 2015. What causes large submarine landslides on low gradient (* 2 degrees) continental slopes with slow (similar to 0.15 m/kyr) sediment accumulation? J. Geophy. Res.-Solid Earth 120, 6722–6739. https://doi.org/10.1002/2015JB012347.

Vanneste M, Forsberg CF, Glimsdal S, et al (2013) Submarine Landslides and Their Consequences: What Do We Know, What Can We Do? In: Margottini C, Canuti P, Sassa K (eds) Landslide Science and Practice: Volume 5: Complex Environment. Springer, Berlin, Heidelberg, pp 5–17.

Verlet, L., 1968. Computer "Experiments" on Classical Fluids. II. Equilibrium Correlation Functions. Phys Rev 165, 201–214. https://doi.org/10.1103/PhysRev.165.201.

Vo, T.-T., Nguyen, T.-K., 2024. Collapse dynamics and deposition morphology of low-viscocohesive granular columns on a rough horizontal surface. Phys. Rev. E 109, 014904. https://doi.org/10.1103/PhysRevE.109.014904.

Vo, T.-T., Nguyen-Thoi, T., 2020. The role of inter-particle friction on rheology and texture of wet granular flows. Eur. Phys. J. E 43, 65. https://doi.org/10.1140/epje/i2020-11987-2.

Yang, G.C., Jing, L., Kwok, C.Y., Sobral, Y.D., 2020. Pore-Scale simulation of Immersed Granular Collapse: Implications to Submarine Landslides. JGR Earth Surface 125, e2019JF005044. https://doi.org/10.1029/2019JF005044.

Yang, C.-F., Chi, W.-C., van Haren, H., 2021a. Deep-sea turbulence evolution observed by multiple closely spaced instruments. Sci. Rep. 11, 3919. https://doi.org/10.1038/s41598-021-83419-2.

Yang, G.C., Jing, L., Kwok, C.Y., Sobral, Y.D., 2021b. Size effects in underwater granular collapses: Experiments and coupled lattice Boltzmann and discrete element method simulations. Phys. Rev. Fluids 6, 114302. https://doi.org/10.1103/PhysRevFluids.6.114302.

Yincan et al Y (2017) Chapter 6 - Submarine Landslides. In: Yincan et al Y (ed) Marine Geo-Hazards in China. Elsevier, pp 179–267.

Zhao, T., Houlsby, G.T., Utili, S., 2014. Investigation of granular batch sedimentation via DEM–CFD coupling. Granul. Matter 16, 921–932. https://doi.org/10.1007/s10035-014-0534-0.

Zhou, G.G.D., Li, S., Song, D., et al., 2019. Depositional mechanisms and morphology of debris flow: physical modelling. Landslides 16, 315–332. https://doi.org/10.1007/s10346-018-1095-9.

Zhu, C., Huang, Y., Sun, J., 2020. Solid-like and liquid-like granular flows on inclined surfaces under vibration – Implications for earthquake-induced landslides. Comput. Geotech. 123, 103598. https://doi.org/10.1016/j.compgeo.2020.103598.

Zhu, R., He, Z., Zhao, K., et al., 2022. Grain-resolving simulations of submerged cohesive granular collapse. J. Fluid Mech. 942, A49. https://doi.org/10.1017/jfm.2022.404.

Zhu, C., Huang, Y., Sun, J., 2023. A granular energy-controlled boundary condition for discrete element simulations of granular flows on erodible surfaces. Comput. Geotech. 154, 105115. https://doi.org/10.1016/j.compgeo.2022.105115.